
Solving Optimal Stopping Problems with Reinforcement Learning

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Abstract

Optimal stopping problems arise throughout finance, operations research and engineering, where one must decide the best time to stop a stochastic process and lock in a reward under uncertainty. When the underlying dynamics are not fully known, reinforcement learning offers a powerful data-driven alternative to the traditional methods that require the model parameters to be specified beforehand. This thesis introduces a unified reinforcement learning framework for several classes of optimal stopping problems, using controlled exploration to regularise the stopping decision and facilitate stable policy iteration. Starting from the entropy-regularised stochastic control problem, we recast optimal stopping as a smooth intensity-based control problem, enabling monotone policy iteration. From there, we apply this reinforcement-learning approach to double stopping problems, where the agent has to determine multiple optimal stopping times based on the same information flow. The main contributions of this thesis arise from our final investigation of time-inconsistent stopping problems, where the optimal decision of the agent can change over time as the discounting factor of the reward is no longer exponential. In discrete time, we formulate this as an entropy-regularised problem with an induced stopping probability. In continuous time, we instead employ a randomised stopping intensity and derive the corresponding weak-equilibrium HJB equation. This thesis also contains numerical experiments, with accompanying code on GitHub¹, to verify the effectiveness of reinforcement learning as a computational tool for complex optimal stopping problems which may not have an analytical solution.

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¹GitHub Link: <https://github.com/chrisjmorton/solving-optimal-stopping-problems-with-RL>

Notation

The notation used throughout this thesis is as follows:

Differential operators and calculus

- $\partial_t = \frac{\partial}{\partial t}$, $\partial_x = \frac{\partial}{\partial x}$, $\partial_{xx} = \frac{\partial^2}{\partial x^2}$.
- $\mathcal{L}$: Infinitesimal generator of diffusion process.
- $C^{1,2}(D)$: Space of functions once continuously differentiable in time and twice in space.
- $L^p(D)$: Lebesgue space with norm $\|u\|_{L^p(D)} = (\int_D |u|^p)^{1/p}$.
- $W_p^{1,2}(D) = \{u \in L^p(D) : \partial_t u, \partial_x u, \partial_{xx} u \in L^p(D)\}$: Parabolic Sobolev space.

Probability spaces and stochastic processes

- $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$: Filtered probability space.
- $\mathbb{X}$: Polish (state) space.
- W_t : Standard Brownian motion.
- X_t, X_t^π : State process.
- $\mathbb{E}, \mathbb{E}_x, \mathbb{E}_{t,x}$: Expectation under $\mathbb{P}$ (unconditional, given $X_0 = x$, given $X_t = x$).
- $\mathcal{B}(E)$: Borel σ -algebra on a topological space E .

Control, policies, and exploration

- U : Action (metric) space.
- T : Terminal time.
- u_t : Classical control process.
- $\mathcal{P}(U)$: Set of probability measures on U .
- $\pi_t(u)$: Control distribution at time t .
- $H(\pi) = - \int_U \pi(u) \log \pi(u) du$: Shannon's differential entropy.
- $R(\pi) = \pi - \pi \log \pi$: Unnormalised entropy.
- $\lambda > 0$: Exploration (entropy regularisation) parameter.

Stopping times and stopping policies

- $\mathcal{T}_{[t,T]}$: Set of $\{\mathcal{F}_t\}$ -stopping times valued in $[t, T]$.
- τ : Stopping time.
- $\pi(t, x)$: Stopping probability/intensity.
- p_t : Survival process.
- $\mathbb{E}_{t,x,p}$: Expectation under $\mathbb{P}$ given $(X_t = x, p_t = p)$.
- $\alpha(x)$: Stopping rate.

Value functions and rewards

- g : Payoff function.
- f : Terminal payoff function.
- $V^u, V^\tau, V^\pi, V^\alpha$: Value functions.
- u_1, u_2, ϕ : Auxiliary value functions in double stopping problems.
- $\delta(t)$: Discounting function.
- r : Discounting rate.

Optimal stopping and equilibria

- V^* : Optimal value function.
- u^* : Optimal control.
- τ^* : Optimal stopping time.
- π^* : Optimal stopping probability/intensity.
- α^* : Optimal equilibrium rate.

1 Introduction & Background

Optimal stopping and stochastic control problems are fundamental in decision-making under uncertainty. They provide a rigorous framework for analysing situations in which an agent observes the evolution of a stochastic system and must decide how to act over time in order to optimise an expected reward. In the classical setting, we model the system dynamics by some stochastic process X and the aim is to determine a set of rules that maximise the expected total reward.² Depending on the problem, these rules can be represented in different ways:

- **Stochastic Control:** The agent can freely adjust a control process u at any time to modify the underlying dynamics and reward function.
- **Optimal Stopping:** The dynamics of X cannot be altered and the agent can only specify a stopping time τ to remove the stochastic component and secure the current reward.

There are several applications to the work developed in this thesis. In mathematical finance, optimal stopping underlies exercise times in option pricing, while stochastic control is useful for more complicated concepts such as optimising a portfolio to achieve risk-adjusted returns. Beyond finance, optimal stopping and stochastic control play important roles in areas such as sequential detection and signal processing in engineering, as well as queuing, inventory management and replacement decisions in operations research.

The theory behind classical stochastic control is well developed in the literature, analysing the problem through the dynamic programming principle and its characterisation as a Hamilton-Jacobi-Bellman (HJB) equation, which reduces optimal decision-making to the study of nonlinear partial differential equations [10] [18] [17] [33]. This framework is then extended to optimal stopping, which can be viewed as a singular control problem where the control collapses to a binary stop/continue decision [22] [5]. The value can then be characterised using variational inequalities or a Snell envelope representation [8]. Multiple stopping extensions have been studied in several forms, including swing options and more general multiple exercise rights [4], and double stopping problems with reduction principles that decompose the two-time optimisation into a sequence of single stopping problems [16] [26] [11]. More recently, time-inconsistent stopping problems, where the discounting factor is no longer exponential, have also been studied using equilibrium notions that replace classical optimality, in both discrete [14] [2] and continuous time [13] [34].

Reinforcement learning (RL) provides a complementary viewpoint by replacing full model specification with learning from simulated or observed trajectories [25] [6] [15]. In contrast to the classical stochastic control paradigm, where the model parameters are known and the optimal strategy is derived from a dynamic programming equation, RL is designed for settings in which the environment is either complex or unknown. In this setup the agent is forced to learn the optimal behaviour through trial and error, interacting with the environment to update its estimates of the future value and refine its strategy over each iteration [25] [15]. This perspective has led to a broad family of model-free methods, including temporal-difference and value-based algorithms such as Q-learning [31] [20] and policy-gradient methods [32].

²In this paper, X is a Markovian process represented by a transition kernel Q in the discrete-time case, or as a stochastic differential equation in the continuous-time case.

Reinforcement learning has increasingly been developed to solve continuous-time stochastic control problems, where exploration can be modelled directly at the level of the controlled dynamics. In this setting exploratory policies are represented as relaxed controls, leading to modified drift and volatility coefficients, as well as an entropy-regularised objective whose value solves an associated HJB equation [29]. This formulation yields an explicit Gaussian optimal exploration policy for linear-quadratic (LQ) models, providing a theoretical justification for using Gaussian exploration in practice [29]. More recently, this control-theoretic RL viewpoint has been extended to optimal stopping by replacing the discontinuous stop/continue decision with a randomised stopping intensity and survival process, transforming the problem into a smooth control setup [7]. This randomised formulation yields a tractable HJB characterisation, a closed-form policy improvement step for the optimal intensity, and a precise convergence-bias trade-off as the exploration parameter varies [7].

In section 2, we first develop an entropy-regularised reinforcement learning framework for optimal stopping through a randomised stopping intensity, leading to a smooth HJB characterisation and monotone policy iteration algorithm. The main contributions of this thesis appear in sections 3, 4, and 5. In particular, we develop a reinforcement learning framework for double stopping problems, applying the classical reduction principle with the entropy-regularised stopping policies developed in [29] to solve the problem. We then further apply this relaxed stopping approach to time-inconsistent stopping problems, where the non-exponential discounting breaks classical optimality and necessitates equilibrium-based solution concepts. We study this in both discrete-time and continuous-time settings, formulating entropy-regularised stopping policies and deriving iterative schemes that converge to equilibrium strategies. Alongside the theoretical developments, this thesis provides reinforcement learning algorithms that can be implemented numerically to solve single, multiple and time-inconsistent stopping problems. These algorithms are designed to be model-free, relying only on simulated or observed trajectories to converge to a solution.

The layout of this thesis is as follows:

- For the remainder of **section 1**, we introduce the classical framework for stochastic control and optimal stopping problems, which underpin the theory developed in this thesis.
- In **section 2**, we draw from [29] and [7] to develop the theory behind entropy-regularised stochastic control, and apply it to optimal stopping by implementing a randomised stopping intensity. We then numerically verify the theory by valuing an American put option.
- In **section 3**, we apply this entropy-regularised framework to double stopping problems, using the reduction principle [16] to convert the formulation into a sequence of single stopping problems. We then numerically verify the theory by valuing an exchange option.
- In **section 4**, we move to time-inconsistent stopping in discrete time with reinforcement learning, employing an entropy-regularised fixed-point iteration method to compute equilibria and solve the problem.
- In **section 5**, we develop the continuous-time analogue for time-inconsistent stopping, using a rate-based randomisation approach to compute equilibria and solve the problem. We then numerically verify the theory by valuing an American put option with hyperbolic discounting.
- Finally, in **section 6** we provide a summary of the results derived in this thesis and suggest directions for future work.

1.1 Classical Stochastic Control

Consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t \geq 0})$ supporting a Brownian motion W_t . The *state dynamics* follow the stochastic differential equation (SDE)

$$dX_t = b(X_t, u_t) dt + \sigma(X_t, u_t) dW_t, \quad X_0 = x, \quad (1)$$

where b and σ are the drift and diffusion coefficients respectively, and u_t is the control process taking values in a metric (action) space U . For this thesis, we assume that the SDE satisfies the following standard requirements to ensure it admits a unique strong solution X :

1. **Measurability:** $b(x, u)$ and $\sigma(x, u)$ are measurable³ functions on $\mathbb{R} \times U$.
2. **Lipschitz continuity:** There exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}$ and $u \in U$.

$$|b(x, u) - b(y, u)| + |\sigma(x, u) - \sigma(y, u)| \leq L|x - y|.$$

3. **Linear growth:** There exists a constant $C > 0$ such that for all $x \in \mathbb{R}$ and $u \in U$,

$$|b(x, u)|^2 + |\sigma(x, u)|^2 \leq C(1 + |x|^2).$$

Given a measurable *reward function* $g : \mathbb{R} \times U \rightarrow \mathbb{R}$, the goal of the stochastic control problem is to determine the optimal action u^* that achieves the maximum expected discounted reward $V^*(t, x)$.

Definition 1.1. A control $u = \{u_s\}_{s \in [0, T]}$ is said to be *admissible* (denoted $u \in \mathcal{A}$) if:

- (i) For all $t \in [0, T]$, u_t is $\{\mathcal{F}_t\}$ -progressively measurable and takes values in U .
- (ii) The state equation admits a unique strong solution X .

Definition 1.2. For any $u \in \mathcal{A}$, the *value function* of the stochastic control problem when following u is

$$V^u(t, x) = \mathbb{E}_{t, x} \left[\int_t^T e^{-r(s-t)} g(X_s, u_s) ds + e^{-r(T-t)} f(X_T) \right], \quad (2)$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is the final payoff and $r > 0$ is the discount rate. In particular, the *optimal value function* is

$$V^*(t, x) = \sup_{u \in \mathcal{A}} \mathbb{E}_{t, x} \left[\int_t^T e^{-r(s-t)} g(X_s, u_s) ds + e^{-r(T-t)} f(X_T) \right], \quad (3)$$

and the *optimal control* is given by

$$u^*(t, x) \in \arg \max_{u \in \mathcal{A}} V^u(t, x). \quad (4)$$

A fundamental tool for analysing these problems is the *dynamic programming principle* (DPP), which expresses the value function recursively in time. In particular, we can use it to derive the HJB equation (PDE), which provides a necessary condition for optimality and reduces the problem to solving a nonlinear partial differential equation. Following [29], this equation can be solved analytically for certain model classes, or numerically using policy iteration techniques.

³In this thesis, measurability refers to Borel measurability with respect to the generated σ -algebra.

Theorem 1.1 (HJB equation). *Suppose $V^* \in C^{1,2}$ and an optimal control u^* exists. Then (u^*, V^*) satisfies the Hamilton-Jacobi-Bellman equation*

$$(\partial_t + \mathcal{L}) V^*(t, x) - rV^*(t, x) + g(x, u^*) = 0, \quad V^*(T, x) = f(x), \quad (5)$$

where

$$\mathcal{L} = \frac{1}{2}\sigma^2(x, u)\partial_{xx} + b(x, u)\partial_x \quad (6)$$

is the infinitesimal generator of X .

Proof. Fix $t \in [0, T]$ and an initial state $x \in \mathbb{R}$ and consider a small time increment $\varepsilon > 0$. For any admissible control $u \in \mathcal{A}$, we have that

$$\begin{aligned} V^*(t, x) &\geq \mathbb{E}_{t,x} \left[\int_t^{t+\varepsilon} e^{-r(s-t)} g(X_s, u_s) ds + e^{-r\varepsilon} V^*(t + \varepsilon, X_{t+\varepsilon}) \right] \\ &= \varepsilon g(x, u) + e^{-r\varepsilon} \mathbb{E}_x [V^*(t + \varepsilon, X_{t+\varepsilon})] + o(\varepsilon), \end{aligned}$$

with equality when $u = u^*$. We can then apply Itô's formula to $V^*(t + \varepsilon, X_{t+\varepsilon})$ to get

$$V^*(t + \varepsilon, X_{t+\varepsilon}) = V^*(t, X_t) + \int_t^{t+\varepsilon} (\partial_t + \mathcal{L}) V^*(s, X_s) ds + \int_t^{t+\varepsilon} \sigma(X_s, u_s) \partial_x V^*(s, X_s) dW_s.$$

After taking expectations, we obtain

$$\begin{aligned} \mathbb{E}_{t,x} [V^*(t + \varepsilon, X_{t+\varepsilon})] &= V^*(t, x) + \mathbb{E}_{t,x} \left[\int_t^{t+\varepsilon} (\partial_t + \mathcal{L}) V^*(s, X_s) ds \right] \\ &= V^*(t, x) + \varepsilon (\partial_t + \mathcal{L}) V^*(t, x) + o(\varepsilon). \end{aligned}$$

Substituting this expression into the original inequality gives

$$\begin{aligned} V^*(t, x) &\geq \varepsilon g(x, u) + e^{-r\varepsilon} (V^*(t, x) + \varepsilon (\partial_t + \mathcal{L}) V^*(t, x)) + o(\varepsilon) \\ &= \varepsilon g(x, u) + (1 - r\varepsilon + o(\varepsilon)) (V^*(t, x) + \varepsilon (\partial_t + \mathcal{L}) V^*(t, x)) + o(\varepsilon) \\ &= V^*(t, x) + \varepsilon g(x, u) - r\varepsilon V^*(t, x) + \varepsilon (\partial_t + \mathcal{L}) V^*(t, x) + o(\varepsilon). \end{aligned}$$

Finally, after cancelling the $V^*(t, x)$ terms, dividing both sides by ε and letting $\varepsilon \rightarrow 0$, we obtain

$$(\partial_t + \mathcal{L}) V^*(t, x) - rV^*(t, x) + g(x, u) \leq 0$$

for any admissible control $u \in \mathcal{A}$. Finally, at the optimal control $u = u^*$, we have that V^* satisfies the HJB equation

$$(\partial_t + \mathcal{L}) V^*(t, x) - rV^*(t, x) + g(x, u^*) = 0$$

as required. $\square$

Remark. The derivation above assumes that the value function V^* is sufficiently smooth ($C^{1,2}$) so that Itô's formula and the DPP can be applied in a classical sense. In general, V^* does not need to be $C^{1,2}$ and the HJB equation is instead understood in the viscosity sense. Under suitable conditions (Lipschitz coefficients, continuous g with at most polynomial growth, compact U), V^* is the unique viscosity solution of the HJB equation. See also [9] and [23].

1.2 Classical Optimal Stopping

In the optimal stopping problem, the control u is now limited to a binary decision: at each time t , the agent either *continues* observing the stochastic process or *stops* the process to receive the current payoff. This reduces the state dynamics of X to:

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad X_0 = x, \quad (7)$$

and leads naturally to the notion of stopping times.

Definition 1.3. A random variable $\tau : \Omega \rightarrow [t, T]$ is called a *stopping time* (with respect to $\{\mathcal{F}_t\}$) if, for every $s \in [t, T]$,

$$\{\tau \leq s\} \in \mathcal{F}_s. \quad (8)$$

We further denote the set of all such stopping times by $\mathcal{T}_{[t, T]}$.

Given a continuous *payoff function* $g : \mathbb{R} \rightarrow \mathbb{R}$, the goal of the optimal stopping problem is to determine the optimal stopping time τ^* that achieves the best expected discounted reward $V^*(x)$.

Definition 1.4. For any stopping time $\tau \in \mathcal{T}_{[t, T]}$, the *value function* of the optimal stopping problem when following τ is defined as

$$V^\tau(t, x) = \mathbb{E}_{t, x} \left[e^{-r(\tau-t)} g(X_\tau) \right], \quad (9)$$

where $r > 0$ is the discount rate. In particular, the *optimal value function* is defined as

$$V^*(t, x) = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}_{t, x} \left[e^{-r(\tau-t)} g(X_\tau) \right], \quad (10)$$

and

$$\tau^*(t, x) \in \arg \max_{\tau \in \mathcal{T}_{[t, T]}} V^\tau(t, x) \quad (11)$$

is called an *optimal stopping time*.

Similarly to subsection 1.1, the DPP yields the following recursive representation of V^* .

Theorem 1.2 (Dynamic Programming Principle). *For any stopping time $\theta \in \mathcal{T}_{[t, T]}$, we have*

$$V^*(t, x) = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}_{t, x} \left[e^{-r(\theta-t)} (g(X_\theta) 1_{\{\tau \leq \theta\}} + V^*(\theta, X_\theta) 1_{\{\tau > \theta\}}) \right]. \quad (12)$$

Intuitive derivation. Fix $\tau \in \mathcal{T}_{[t, T]}$ and split Ω into $\{\tau \leq \theta\}$ and $\{\tau > \theta\}$. Using the tower property,

$$\begin{aligned} \mathbb{E}_{t, x} [e^{-r(\tau-t)} g(X_\tau)] &= \mathbb{E}_{t, x} \left[e^{-r(\tau-t)} g(X_\tau) 1_{\{\tau \leq \theta\}} + e^{-r(\tau-t)} g(X_\tau) 1_{\{\tau > \theta\}} \right] \\ &= \mathbb{E}_{t, x} \left[e^{-r(\tau-t)} g(X_\tau) 1_{\{\tau \leq \theta\}} + e^{-r(\theta-t)} \mathbb{E} \left[e^{-r(\tau-\theta)} g(X_\tau) \mid \mathcal{F}_\theta \right] 1_{\{\tau > \theta\}} \right]. \end{aligned}$$

Since the continuation value is bounded from above by $V^*(\theta, X_\theta)$, taking the essential supremum over τ yields (12). $\square$

A key consequence is that the optimal stopping time τ^* can be expressed as the first time that the current payoff g exceeds the value V^* . Mathematically, this can be represented as:

$$\tau^*(t, x) = \inf\{s \geq t : V^*(s, X_s) = g(X_s)\} \wedge T. \quad (13)$$

Under the same regularity conditions imposed in subsection 1.1, the value function admits a PDE characterisation analogous to the HJB equation.

Theorem 1.3 (Variational inequality). *Suppose that $V^* \in C^{1,2}$. Then V^* satisfies*

$$\max\{(\partial_t + \mathcal{L})V^*(t, x) - rV^*(t, x), g(x) - V^*(t, x)\} = 0, \quad V^*(T, x) = g(x), \quad (14)$$

where

$$\mathcal{L} = \frac{1}{2}\sigma^2(x)\partial_{xx} + b(x)\partial_x \quad (15)$$

is the infinitesimal generator of X .

Intuitive derivation. Fix (t, x) and consider a small time increment $\varepsilon > 0$. By the DPP, delaying stopping yields

$$V^*(t, x) \geq \mathbb{E}_{t,x} [e^{-r\varepsilon} V^*(t + \varepsilon, X_{t+\varepsilon})],$$

with equality if it is optimal to continue. We then apply Itô's formula on $V^*(t + \varepsilon, X_{t+\varepsilon})$ to get:

$$V^*(t + \varepsilon, X_{t+\varepsilon}) = V^*(t, x) + \int_t^{t+\varepsilon} (\partial_t + \mathcal{L})V^*(s, X_s) ds + \int_t^{t+\varepsilon} \sigma(X_s)\partial_x V^*(s, X_s) dW_s.$$

After taking expectations, we obtain:

$$\mathbb{E}_{t,x}[V^*(t + \varepsilon, X_{t+\varepsilon})] = V^*(t, x) + \varepsilon (\partial_t + \mathcal{L})V^*(t, x) + o(\varepsilon).$$

Substituting this expression into the dynamic programming inequality gives:

$$\begin{aligned} V^*(t, x) &\geq e^{-r\varepsilon} [V^*(t, x) + \varepsilon (\partial_t + \mathcal{L})V^*(t, x)] + o(\varepsilon) \\ &= (1 - r\varepsilon + o(\varepsilon)) [V^*(t, x) + \varepsilon (\partial_t + \mathcal{L})V^*(t, x)] + o(\varepsilon) \\ &= V^*(t, x) + \varepsilon ((\partial_t + \mathcal{L})V^*(t, x) - rV^*(t, x)) + o(\varepsilon). \end{aligned}$$

Finally, after cancelling the $V^*(t, x)$ terms, dividing by ε , and letting $\varepsilon \rightarrow 0$, we obtain

$$(\partial_t + \mathcal{L})V^*(t, x) - rV^*(t, x) \leq 0,$$

with equality if it is optimal to continue at (t, x) . On the other hand, if it is optimal to stop, then

$$V^*(t, x) = g(x)$$

with $V^*(t, x) > g(x)$ otherwise. This yields the variational inequality satisfied by V^* . $\square$

Remark. Similarly to Theorem 1.1, V^* does not need to be smooth and the variational inequality is instead interpreted in the viscosity sense. Under the same conditions, V^* is the unique viscosity solution of the associated variational inequality. See also [23].

2 Reinforcement Learning

Classical solutions assume full knowledge of model parameters and rely on dynamic programming and PDEs [29] [7]. In practice, these parameters are often unknown, making these approaches infeasible. This motivates *reinforcement learning* (RL), which instead learns optimal strategies through interaction with the environment and data. It does this by striking a balance between:

- **Exploitation:** Choosing actions that maximise rewards based on the agent’s current knowledge of the model.
- **Exploration:** Trying new actions to gather information about the model for future gains.

For this section, we once again assume the underlying stochastic process X is governed by the SDE given in (1) for stochastic control and (7) for optimal stopping, but the model parameters b and σ are now unknown. We will work with a finite time horizon ($T < \infty$), and assume we can attain realisations of X to learn the model parameters and identify the optimal decision. Once again, the goal is to find the optimal value function V^* and corresponding control/stopping time, as defined in (3) and (10) for stochastic control and optimal stopping respectively.

2.1 Relaxed Stochastic Control Formulation

For reinforcement learning in stochastic control, the key idea is that we model exploration using a *distribution* of controls $\pi = \{\pi_t(u)\}_{t \geq 0}$.

Definition 2.1. A *relaxed control* is a family $\pi = \{\pi_t(u)\}_{t \geq 0}$ such that for each t , $\pi_t \in \mathcal{P}(U)$ is a progressively measurable probability measure on the action space U .⁴

This allows us to explore and interact with the environment to learn its unknown dynamics through trial and error. At each learning step, the agent samples N classical controls from the distribution π and evaluates the policy by averaging the observed rewards. This procedure is one of the most fundamental components of reinforcement learning algorithms [25], and becomes more accurate as N increases. It is thus important for us to consider the limit as $N \rightarrow \infty$. From the law of large numbers, we have that

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \Delta x_t^i &= \frac{1}{N} \sum_{i=1}^N (x_{t+\Delta t}^i - x_t^i) \\ &\approx \frac{1}{N} \sum_{i=1}^N b(x_t^i, u_t^i) \Delta t + \frac{1}{N} \sum_{i=1}^N \sigma(x_t^i, u_t^i) (W_{t+\Delta t}^i - W_t^i) \\ &\longrightarrow \mathbb{E} \left[\int_U b(X_t^\pi, u) \pi_t(u) du \Delta t \right] + \mathbb{E} \left[\int_U \sigma(X_t^\pi, u) \pi_t(u) du \right] \mathbb{E} [W_{t+\Delta t} - W_t] \\ &= \mathbb{E} \left[\int_U b(X_t^\pi, u) \pi_t(u) du \Delta t \right]. \end{aligned}$$

⁴In this context, we can treat any classical control $u = \{u_t\}_{t \geq 0}$ as a Dirac measure ($\pi_t(\cdot) = \delta_{u_t}(\cdot)$), making the formulation an extension of the classical theory.

Similarly, looking at the mean square we get

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \Delta(x_t^i)^2 &\approx \frac{1}{N} \sum_{i=1}^N \sigma^2(x_t^i, u_t^i) \Delta t \\ &\longrightarrow \mathbb{E} \left[\int_U \sigma^2(X_t^\pi, u) \pi_t(u) du \Delta t \right]. \end{aligned}$$

This interpretation leads us to the *exploratory formulation* of the controlled state dynamics.

Definition 2.2. Given a relaxed control π , the *exploratory dynamics* of X_t^π follow

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t) dt + \tilde{\sigma}(X_t^\pi, \pi_t) dW_t, \quad X_0^\pi = x, \quad (16)$$

where

$$\tilde{b}(x, \pi) = \int_U b(x, u) \pi(u) du, \quad \tilde{\sigma}^2(x, \pi) = \int_U \sigma^2(x, u) \pi(u) du.$$

For the reward, we have

$$\frac{1}{N} \sum_{i=1}^N e^{-rt} g(x_t^i, u_t^i) \Delta t \longrightarrow \mathbb{E} \left[e^{-rt} \int_U g(X_t^\pi, u) \pi_t(u) du \Delta t \right] \quad (17)$$

almost surely by the law of large numbers. This suggests we should further update the reward function g to the *exploratory reward* $\tilde{g}(y, \pi) = \int_U g(y, u) \pi(u) du$.

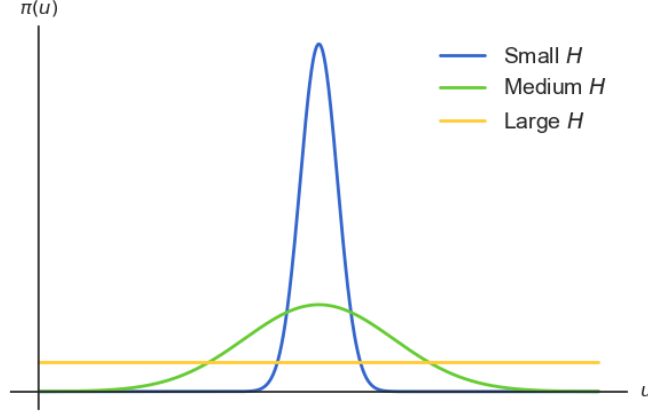
Definition 2.3. We say that a relaxed control π is *admissible* (denoted $\pi \in \mathcal{A}$) if:

- (i) For any $t \in [0, T]$, $\pi_t \in \mathcal{P}(U)$.
- (ii) For any $A \in \mathcal{B}(U)$, $\{\int_A \pi_t(u) du\}_{t \geq 0}$ is $\mathcal{F}_t$ -progressively measurable.
- (iii) The stochastic differential equation (16) has a unique strong solution X^π if π is applied.
- (iv) $V^*(t, x) < \infty$.

To promote exploration and account for uncertainty in the controlled state dynamics, we add Shannon's [24] differential entropy

$$H(\pi_t) = - \int_U \pi_t(u) \log \pi_t(u) du \quad (18)$$

as a regularisation term. Figure 1 shows a plot of three distributions with varying entropy. Intuitively, this method incentivises uncertainty about the value of the sampled controls, punishing the objective function if it relies too heavily on one choice without testing alternative strategies. If we were to use a classical control (Dirac distribution) and follow a pure exploitation method, then $H(\pi_t) \rightarrow -\infty$. This technique is known as entropy-regularisation, and is widely used in the field to encourage exploration [1] [12] [21].


 Figure 1: Plot of distributions with varying $H(\pi)$.

Definition 2.4. The *value function* of the *entropy-regularised relaxed stochastic control problem* under $\pi \in \mathcal{A}$ is given by

$$\begin{aligned} V^\pi(t, x) &= \mathbb{E}_{t,x} \left[\int_t^T e^{-r(s-t)} \left(\int_U g(X_s^\pi, u) \pi_s(u) du + \lambda H(\pi_s) \right) ds + e^{-r(T-t)} f(X_T^\pi) \right] \\ &= \mathbb{E}_{t,x} \left[\int_t^T e^{-r(s-t)} \left(\tilde{g}(X_s^\pi, \pi_s) + \lambda H(\pi_s) \right) ds + e^{-r(T-t)} f(X_T^\pi) \right] \end{aligned} \quad (19)$$

where $\lambda > 0$ is the *exploration weight parameter*. In particular, the *optimal value function* is

$$V^*(t, x) = \mathbb{E}_{t,x} \left[\int_t^T e^{-r(s-t)} \left(\tilde{g}(X_s^\pi, \pi_s) + \lambda H(\pi_s) \right) ds + e^{-r(T-t)} f(X_T^\pi) \right]. \quad (20)$$

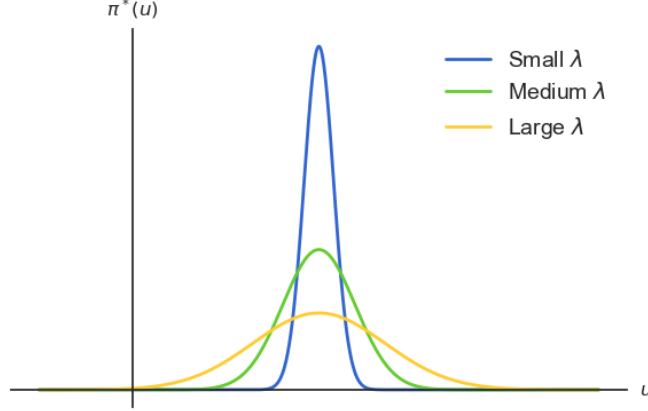
The associated HJB equation is

$$rV^*(t, x) = \sup_{\pi \in \mathcal{A}} \int_U \left(g(x, u) - \lambda \log \pi_t(u) + \frac{1}{2} \sigma^2(x, u) \partial_{xx} V^*(t, x) + b(x, u) \partial_x V^*(t, x) \right) \pi_t(u) du, \quad (21)$$

with $V^*(T, x) = f(x)$. Maximising this gives us the optimal control

$$\pi_t^*(u) = \frac{\exp \left(\frac{1}{\lambda} [g(x, u) + \frac{1}{2} \sigma^2(x, u) \partial_{xx} V^*(t, x) + b(x, u) \partial_x V^*(t, x)] \right)}{\int_U \exp \left(\frac{1}{\lambda} [g(x, \tilde{u}) + \frac{1}{2} \sigma^2(x, \tilde{u}) \partial_{xx} V^*(t, x) + b(x, \tilde{u}) \partial_x V^*(t, x)] \right) d\tilde{u}}. \quad (22)$$

A plot of π^* for varying values of λ is shown in Figure 2. Once again, we see that a larger λ encourages more exploration/uncertainty in the optimal solution, while a smaller λ focuses more on exploitation. A sketch of the proof for the optimal control formula can originally be found in [29], however we provide more details here. Note that here we assume that $V \in C^{1,2}$ satisfies the standard regularity conditions.


 Figure 2: Plot of π^* of an example problem for varying λ .

Theorem 2.1 (HJB and softmax feedback distribution). *Let V^* be the value in (20) under the exploratory dynamics (16). Then the corresponding HJB equation of the problem is given by (21) and any sufficiently smooth solution to the equation admits the pointwise maximiser given in (22).*

Proof. We will derive the HJB and the corresponding maximiser for π in two separate steps.

Step 1: DPP and first-order expansion. Fix x and a small $\varepsilon > 0$. By the DPP,

$$V^*(t, x) = \sup_{\pi \in \mathcal{A}} \mathbb{E}_{t,x} \left[\int_t^{t+\varepsilon} e^{-r(s-t)} (\tilde{g}(X_s, \pi_s) + \lambda H(\pi_s)) ds + e^{-r\varepsilon} V^*(t + \varepsilon, X_{t+\varepsilon}) \right].$$

Now, using the Itô formula on $V^*(t + \varepsilon, X_{t+\varepsilon})$ and taking expectations, we get

$$\mathbb{E}_{t,x} [V^*(t + \varepsilon, X_{t+\varepsilon})] = V^*(t, x) + \mathbb{E}_{t,x} \left[\int_t^{t+\varepsilon} \left(\tilde{b}(X_s, \pi_s) \partial_x V^*(s, X_s) + \frac{1}{2} \tilde{\sigma}^2(X_s, \pi_s) \partial_{xx} V^*(s, X_s) \right) ds \right].$$

Substituting this into the DPP equation and using the expansion $e^{-r\varepsilon} = 1 - r\varepsilon + o(\varepsilon)$ gives us

$$V^*(t, x) = \sup_{\pi \in \mathcal{A}} \left\{ \varepsilon (\tilde{g}(x, \pi_t) + \lambda H(\pi_t)) + (1 - r\varepsilon) \left(V^*(t, x) + \varepsilon (\tilde{b}(x, \pi_t) \partial_x V^*(t, x) + \frac{1}{2} \tilde{\sigma}^2(x, \pi_t) \partial_{xx} V^*(t, x)) \right) \right\} + o(\varepsilon).$$

Now, if we cancel $V^*(t, x)$ on both sides and divide by ε we have

$$0 = \sup_{\pi \in \mathcal{A}} \left\{ \tilde{g}(x, \pi_t) + \lambda H(\pi_t) + \tilde{b}(x, \pi_t) \partial_x V^*(t, x) + \frac{1}{2} \tilde{\sigma}^2(x, \pi_t) \partial_{xx} V^*(t, x) - r V^*(t, x) \right\} + o(1),$$

which after letting $\varepsilon \downarrow 0$ and rearranging gives us the HJB

$$0 = \sup_{\pi \in \mathcal{A}} \left\{ \tilde{g}(x, \pi_t) + \lambda H(\pi_t) + \tilde{b}(x, \pi_t) \partial_x V^*(t, x) + \frac{1}{2} \tilde{\sigma}^2(x, \pi_t) \partial_{xx} V^*(t, x) - r V^*(t, x) \right\},$$

or, equivalently,

$$r V^*(t, x) = \sup_{\pi \in \mathcal{A}} \int_U \left(g(x, u) - \lambda \log \pi_t(u) + b(x, u) \partial_x V^*(t, x) + \frac{1}{2} \sigma^2(x, u) \partial_{xx} V^*(t, x) \right) \pi_t(u) du.$$

Step 2: Pointwise maximisation. For this part, we follow a standard Lagrange multiplier approach. For a fixed x , define $\Phi(x, u) = g(x, u) + b(x, u)\partial_x V^*(t, x) + \frac{1}{2}\sigma^2(x, u)\partial_{xx} V^*(t, x)$.

For this step, we now must maximise

$$\sup_{\pi \in \mathcal{A}} \int_U (\Phi(x, u) - \lambda \log \pi_t(u)) \pi_t(u) du$$

such that $\int_U \pi_t(u) du = 1$. The Lagrangian for this problem is

$$\mathcal{L}(\pi_t, \eta) = \int_U \Phi(x, u) \pi_t(u) du - \lambda \int_U \pi_t(u) \log \pi_t(u) du + \eta \left(1 - \int_U \pi_t(u) du \right).$$

Applying the first-order condition (Gâteaux derivative) gives us that for any perturbation $\delta\pi_t$ with $\int_U \delta\pi_t du = 0$,

$$0 = \frac{d}{d\varepsilon} \mathcal{L}(\pi_t + \varepsilon \delta\pi_t, \eta) \Big|_{\varepsilon=0} = \int_U (\Phi(x, u) - \lambda(1 + \log \pi_t(u)) - \eta) \delta\pi_t(u) du.$$

This means that the bracket must vanish, giving us

$$\Phi(x, u) - \lambda(1 + \log \pi_t(u)) - \eta = 0 \implies \log \pi_t(u) = \lambda^{-1}(\Phi(x, u) - \eta - \lambda).$$

Finally, if we exponentiate both sides and normalise to ensure $\int_U \pi du = 1$, we get

$$\begin{aligned} \pi_t^*(u) &= \frac{\exp\{\Phi(x, u)/\lambda\}}{\int_U \exp\{\Phi(x, \tilde{u})/\lambda\} d\tilde{u}} \\ &= \frac{\exp\left(\frac{1}{\lambda} \left[g(x, u) + \frac{1}{2}\sigma^2(x, u)\partial_{xx} V^*(t, x) + b(x, u)\partial_x V^*(t, x) \right]\right)}{\int_U \exp\left(\frac{1}{\lambda} \left[g(x, \tilde{u}) + \frac{1}{2}\sigma^2(x, \tilde{u})\partial_{xx} V^*(t, x) + b(x, \tilde{u})\partial_x V^*(t, x) \right]\right) d\tilde{u}} \end{aligned}$$

as required. This is the continuous-time analogue of the softmax rule and matches [29]. $\square$

Remark. The derivation above assumes that the value function V^* is sufficiently smooth so that the DPP and the pointwise maximisation over control distributions can be carried out formally. A fully rigorous proof requires consideration of the well-posedness of the entropy-regularised HJB equation, a verification theorem establishing optimality of the feedback control, and sufficient regularity of the problem to justify the use of the DPP. Since the focus of this thesis is on optimal stopping and how to transform them into reinforcement learning problems, we do not reproduce these arguments here and instead refer the reader to [27] for full details.

2.2 Application to Optimal Stopping

For optimal stopping problems, our previous reinforcement learning method now leads to issues as we have a non-smooth optimisation problem. Similarly to subsection 1.2, we work with a finite time horizon ($T < \infty$) and assume the dynamics of X follow (7), however we are required to learn the model parameters from data. Following the exploratory stopping framework of [7], we instead use a *stopping intensity* $\pi_s = \pi(s, X_s) \geq 0$ to solve for V^* originally defined in (10).

Definition 2.5. For any $t \in [0, T]$, a *stopping intensity* is a measurable function $\pi : [t, T] \times \mathbb{R} \rightarrow \mathbb{R}$ that satisfies

$$\pi(s, x) \geq 0$$

for all $(s, x) \in [t, T] \times \mathbb{R}$. Moreover, for any $s \in [t, T]$, the associated *survival process* p satisfies

$$dp_s = -\pi_s p_s ds, \quad p_t = 1, \quad (23)$$

representing the probability of not stopping before time s .

From this definition, $\pi_s ds$ can be seen as the probability of the agent stopping within $[s, s + ds]$, given that they have not stopped before s , while $\pi_s p_s ds$ represents the unconditional probability of stopping on $[s, s + ds]$. A plot of an example stopping intensity and corresponding survival process is shown in Figure 3. Since we are now dealing with a stopping intensity, we also change the exploration regularisation term to the unnormalised entropy $R(\pi) = \pi - \pi \log \pi$ [35], which is plotted in Figure 4. This leads us to the entropy-regularised optimal stopping problem.

Definition 2.6. Given a stopping intensity π , we define the *entropy-regularised value function* $V^\pi(t, x) = V^\pi(t, x; \lambda)$ as

$$V^\pi(t, x) = \mathbb{E} \left[\int_t^T e^{-r(s-t)} \left(g(X_s) \pi(s, X_s) p_s + \lambda R(\pi(s, X_s)) p_s \right) ds + e^{-r(T-t)} g(X_T) p_T \right], \quad (24)$$

where $\lambda > 0$ is the *exploration parameter*.

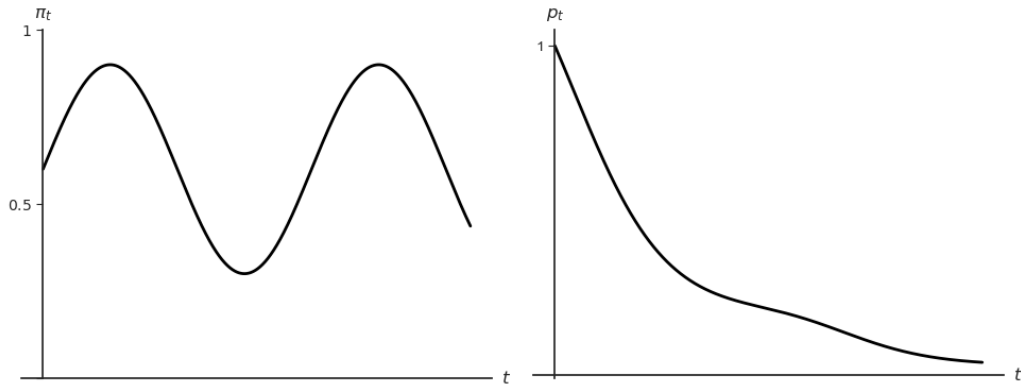
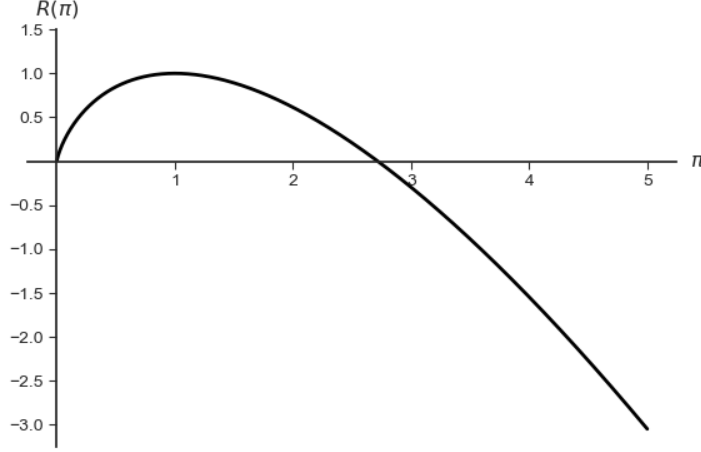


Figure 3: Example stopping intensity (left) and corresponding survival process (right) over time.


 Figure 4: Plot of $R(\pi)$.

In the classical optimal stopping problem, the value function is characterised by the variational inequality $\max \{(\partial_t + \mathcal{L})V^* - rV^* = 0, g - V^*\} = 0$, meaning the exercise boundary is typically non-smooth and the optimisation over a binary action creates a spike at $V^* = g$. In this relaxed formulation, the stop/continue decision is now encoded through a hazard $\pi(t, X_t)$ and survival process p_t , and the optimal HJB equation for V^* includes an exponential term that becomes increasingly steep as $\lambda \downarrow 0$. In doing so, the optimal stopping problem is transformed into a family of smooth stochastic control problems, whose solutions converge back to the classical value as the exploration parameter vanishes. This means we can now apply reinforcement learning to solve the optimal stopping problem.

The derivation of the HJB equation for (24) as well as the optimal stopping intensity π^* was originally shown in [7], however we will provide more details in the derivation here. To complete the proof, we will define the augmented value function

$$\tilde{V}^\pi(t, x, p) = \mathbb{E}_{t,x,p} \left[\int_t^T e^{-r(s-t)} \left(g(X_s) \pi_s p_s + \lambda R(\pi_s) p_s \right) ds + e^{-r(T-t)} g(X_T) p_T \right], \quad (25)$$

where we have relaxed p to no longer require $p_t = 1$. We also note that the proof of existence and uniqueness of a classical solution $V^\pi \in C^{1,2}([0, T] \times \mathbb{R})$ follows the standard parabolic theory for semi-linear equations with bounded terminal conditions, and refer the reader to [10] for more information.

Note that if $t = 0$ and $p_0 = 1$, then the solution of (23) is $p_s = \exp(-\int_0^s \alpha(X_r) dr)$, meaning that $\tilde{V}^\pi(0, x, 1) = V^\pi(0, x)$. Extending on this, we further have the relationship

$$V^\pi(t, x) = \tilde{V}^\pi(t, x, 1)$$

for any (t, x) , as well as the property that

$$pV^\pi(t, x) = \tilde{V}^\pi(t, x, p)$$

for any (t, x, p) .

Theorem 2.2 (HJB equation, optimal intensity, well-posedness). *For any fixed π , the value $V^\pi(t, x)$ defined in (24) is the unique $C^{1,2}$ solution to the HJB equation*

$$(\partial_t + \mathcal{L})V^\pi(t, x) - \alpha V^\pi(t, x) + (g - V^\pi(t, x))\pi(t, x) + \lambda R(\pi(t, x)) = 0, \quad V^\pi(T, x) = g(x) \quad (26)$$

where $\mathcal{L} = \frac{1}{2}\sigma^2(x)\partial_{xx} + b(x)\partial_x$.

In particular, the optimal value function $V^*(t, x) = V^{\pi^*}(t, x) = \sup_{\pi \geq 0} V^\pi(t, x)$ satisfies the HJB equation

$$(\partial_t + \mathcal{L})V^*(t, x) - \alpha V^*(t, x) + \lambda \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right) = 0, \quad V^*(T, x) = g(x), \quad (27)$$

and uses the pointwise maximiser

$$\pi^*(t, x) = \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right). \quad (28)$$

Moreover, for any admissible π ,

$$V^\pi(t, x) \leq \|g\|_\infty + \lambda(T - t), \quad 0 \leq t \leq T.$$

Proof. We will break the proof into three steps.

Step 1: HJB equation. Fix (t, x, p) and $\varepsilon > 0$. By the DPP,

$$\begin{aligned} \tilde{V}^\pi(t, x, p) &= \mathbb{E}_{t,x,p} \left[\int_t^{t+\varepsilon} e^{-r(s-t)} (g(X_s)\pi_s p_s + \lambda R(\pi_s)p_s) ds + e^{-r\varepsilon} \tilde{V}^\pi(t + \varepsilon, X_{t+\varepsilon}, p_{t+\varepsilon}) \right] \\ &= \varepsilon (g(x)\pi(t, x)p + \lambda R(\pi(t, x))p) + (1 - \alpha\varepsilon) \mathbb{E}_{t,x,p} [\tilde{V}^\pi(t + \varepsilon, X_{t+\varepsilon}, p_{t+\varepsilon})] + o(\varepsilon). \end{aligned}$$

We can now apply Itô's formula on $p_s V^\pi(s, X_s)$ on $[t, t + \varepsilon]$ (since p_s has finite variation) to get:

$$\mathbb{E}_{t,x,p} [p_{t+\varepsilon} V^\pi(t + \varepsilon, X_{t+\varepsilon})] = p V^\pi(t, x) + \varepsilon p \left((\partial_t + \mathcal{L})V^\pi(t, x) - \pi(t, x)V^\pi(t, x) \right) + o(\varepsilon).$$

Substituting this into the DPP expansion and cancelling $\tilde{V}^\pi(t, x, p) = p V^\pi(t, x)$ yields

$$0 = \varepsilon p \left((\partial_t + \mathcal{L})V^\pi(t, x) - \pi(t, x)V^\pi(t, x) - \alpha V^\pi(t, x) \right) + \varepsilon p (g(x)\pi(t, x) + \lambda R(\pi(t, x))) + o(\varepsilon).$$

Dividing by $\varepsilon p > 0$ and letting $\varepsilon \rightarrow 0$ gives

$$(\partial_t + \mathcal{L})V^\pi(t, x) - \alpha V^\pi(t, x) + (g(x) - V^\pi(t, x))\pi(t, x) + \lambda R(\pi(t, x)) = 0, \quad V^\pi(T, x) = g(x).$$

In particular, taking the supremum over $\pi \geq 0$ yields

$$(\partial_t + \mathcal{L})V^*(t, x) - \alpha V^*(t, x) + \sup_{\pi \geq 0} \left[(g(x) - V^*(t, x))\pi(t, x) + \lambda R(\pi(t, x)) \right] = 0,$$

with terminal condition $V^*(T, x) = g(x)$.

Step 2: π^* identification. We now determine $\pi^*(t, x)$ by finding the supremum. For any fixed (t, x) , define

$$H(x, \pi, V^*) = (g(x) - V^*(t, x))\pi + \lambda R(\pi).$$

Since

$$\partial_\pi H(x, \pi, V^*) = g(x) - V^*(t, x) - \lambda \log \pi, \quad \partial_{\pi\pi} H(x, \pi, V^*) = -\frac{\lambda}{\pi} < 0,$$

the maximiser π^* is unique. Setting $\partial_\pi H(x, \pi, V^*) = 0$ gives

$$\pi^*(t, x) = \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right).$$

Evaluating H at π^* yields

$$\sup_{\pi \geq 0} H(x, \pi, V^*) = H(x, \pi^*, V^*) = \lambda \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right),$$

and thus V^* satisfies

$$(\partial_t + \mathcal{L})V^*(t, x) - \alpha V^*(t, x) + \lambda \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right) = 0, \quad V^*(T, x) = g(x).$$

Step 3: Well-posedness and bounds. Fix any admissible π . Consider the admissible control $\pi(t, \cdot) \equiv 0$. Then $p_t \equiv 1$, and

$$V^\pi(t, x) \leq \mathbb{E}\left[\int_t^T e^{-r(s-t)} \lambda ds + e^{-r(T-t)} g(X_T)\right] \leq \lambda(T-t) + \|g\|_\infty.$$

This completes the proof. $\square$

It was further found in [7] that the bias to the classical problem is $\mathcal{O}(\lambda \log(1/\lambda))$. Moreover, the entropy-regularised value and induced stopping time converges to the classical solution as $\lambda \downarrow 0$.

Theorem 2.3 (Bias and convergence as $\lambda \downarrow 0$). *Let $V^\pi(t, x; \lambda)$ denote the value function of the intensity-regularised optimal stopping problem in (24), and let*

$$\widehat{V}^*(t, x) = \sup_{\tau \in \mathcal{T}[t, T]} \mathbb{E}_{t, x} \left[e^{-r(\tau-t)} g(X_\tau) \right]$$

be the value function of the classical optimal stopping problem. Then there exists a constant $C > 0$ independent of λ such that for all $(t, x) \in [0, T] \times \mathbb{R}$ and all sufficiently small $\lambda > 0$,

$$V^\pi(t, x; \lambda) - \lambda \log(1 + T - t) \leq \widehat{V}^*(t, x) \leq V^\pi(t, x; \lambda) + \lambda \log\left(\frac{C}{\lambda}\right).$$

In particular,

$$\|V^\pi - \widehat{V}^*\|_\infty = \mathcal{O}\left(\lambda \log \frac{1}{\lambda}\right)$$

as $\lambda \downarrow 0$.

Proof. We will split the proof into three steps.

Step 1: Lower bound above the obstacle.

Recall that V^λ satisfies the penalised HJB equation

$$(\partial_t + \mathcal{L})V^\pi(t, x; \lambda) - rV^\pi(t, x; \lambda) + \lambda \exp\left(-\frac{V^\pi(t, x; \lambda) - g(x)}{\lambda}\right) = 0, \quad V^\pi(T, x; \lambda) = g(x).$$

We claim that there exists a constant $C > 0$ independent of λ such that

$$V^\pi(t, x; \lambda) \geq g(x) - \lambda \log\left(\frac{C}{\lambda}\right) \quad \text{for all } (t, x) \in [0, T] \times \mathbb{R}.$$

Suppose for contradiction that this bound fails. Then for arbitrarily small $\lambda > 0$ there exists (t_0, x_0) such that

$$V^\pi(t_0, x_0; \lambda) - g(x_0) < -\lambda \log\left(\frac{C}{\lambda}\right).$$

Fix $\varepsilon > 0$ and define the auxiliary function

$$\Phi(t, x) = V^\pi(t, x; \lambda) - g(x) + \lambda \log\left(\frac{C}{\lambda}\right) - \varepsilon(|x|^2 + T - t).$$

For C fixed and λ small, Φ attains a strict negative minimum at some interior point $(t^*, x^*) \in (0, T) \times \mathbb{R}$. At this point,

$$\partial_t \Phi(t^*, x^*) \leq 0, \quad \partial_x \Phi(t^*, x^*) = 0, \quad \partial_{xx} \Phi(t^*, x^*) \geq 0.$$

Applying the parabolic operator to $V^\lambda - g$ and using the above inequalities, we obtain at (t^*, x^*)

$$(\partial_t + \mathcal{L})(V^\pi - g) \geq -C_1 \varepsilon,$$

where $C_1 > 0$ is independent of λ . On the other hand, subtracting the equation for g from the HJB satisfied by V^π gives

$$(\partial_t + \mathcal{L})(V^\pi - g) = r(V^\pi - g) - \lambda \exp\left(-\frac{V^\pi - g}{\lambda}\right).$$

At the minimum point we have

$$\lambda \exp\left(-\frac{V^\pi - g}{\lambda}\right) \geq \lambda \frac{C}{\lambda} = C,$$

and hence

$$(\partial_t + \mathcal{L})(V^\pi - g) \leq r(V^\pi - g) - C \leq -C.$$

Combining the two estimates yields

$$-C \geq -C_1\varepsilon,$$

which is impossible for ε sufficiently small. This contradiction proves the stated lower bound.

Step 2: Upper bound $\widehat{V}^* \leq V^\pi + O(\lambda \log \lambda)$.

Suppose, for sake of contradiction, that

$$\sup_{t,x} \left(\widehat{V}^*(t, x) - V^\pi(t, x; \lambda) - \lambda \log(C/\lambda) \right) > 0.$$

Fix $\varepsilon > 0$ and define

$$w(t, x; \lambda) = V^\pi(t, x; \lambda) + \lambda \log\left(\frac{C}{\lambda}\right) + \varepsilon(|x|^2 + T - t).$$

Since $w(t, x; \lambda) \rightarrow +\infty$ as $|x| \rightarrow \infty$ or $t \downarrow 0$, the function $\widehat{V}^* - w$ attains a strict positive maximum at some $(t^*, x^*) \in (0, T) \times \mathbb{R}$.

At this point we must have $\widehat{V}^*(t^*, x^*) > g(x^*)$, since otherwise $\widehat{V}^*(t^*, x^*) - w(t^*, x^*; \lambda) < 0$. Hence (t^*, x^*) lies in the continuation region of the classical problem, and by the smooth fit principle $\widehat{V}^*$ is $C^{1,2}$ locally and satisfies

$$(\partial_t + \mathcal{L})\widehat{V}^*(t^*, x^*) - r\widehat{V}^*(t^*, x^*) = 0.$$

At the strict maximum of $\widehat{V}^* - w$ we have

$$\partial_t(\widehat{V}^* - w) = 0, \quad \partial_x(\widehat{V}^* - w) = 0, \quad \partial_{xx}(\widehat{V}^* - w) \leq 0,$$

which implies

$$(\partial_t + \mathcal{L})\widehat{V}^* \leq (\partial_t + \mathcal{L})V^\pi + O(\varepsilon).$$

Subtracting $r\widehat{V}^* = rV^\pi + O(\varepsilon)$ yields

$$(\partial_t + \mathcal{L})V^\pi(t^*, x^*; \lambda) - rV^\pi(t^*, x^*; \lambda) \geq -O(\varepsilon).$$

However, the HJB equation for V^π gives

$$(\partial_t + \mathcal{L})V^\pi - rV^\pi = -\lambda \exp\left(-\frac{V^\pi - g}{\lambda}\right).$$

By Step 1, the exponential term is uniformly bounded below by a positive constant, so the right hand side is bounded away from zero independently of ε . This yields a contradiction for ε sufficiently small.

Therefore,

$$\widehat{V}^*(t, x) \leq V^\pi(t, x; \lambda) + \lambda \log\left(\frac{C}{\lambda}\right).$$

Step 3: Lower bound $V^\pi \leq \widehat{V}^* + \lambda \log(1 + T - t)$.

Define $f(t) = \log(1 + T - t)$ (which satisfies $\partial_t f + e^{-f} = 0$), as well as

$$\bar{V}(t, x) = \widehat{V}^*(t, x) + \lambda f(t).$$

Since $\widehat{V}^* \geq g$, we have

$$\lambda \exp\left(-\frac{\bar{V} - g}{\lambda}\right) \leq \lambda e^{-f(t)}.$$

Combining this with the fact that $\widehat{V}^*$ is the optimal solution to the HJB equation (27) gives us

$$(\partial_t + \mathcal{L})\bar{V} - r\bar{V} + \lambda \exp\left(-\frac{\bar{V} - g}{\lambda}\right) \leq 0.$$

By the comparison principle, this means we have

$$V^\pi(t, x; \lambda) \leq \bar{V}(t, x),$$

that is,

$$V^\pi(t, x; \lambda) - \lambda \log(1 + T - t) \leq \widehat{V}^*(t, x).$$

Now, combining Steps 2 and 3 gives us

$$V^\pi(t, x; \lambda) - \lambda \log(1 + T - t) \leq \widehat{V}^*(t, x) \leq V^\pi(t, x; \lambda) + \lambda \log\left(\frac{C}{\lambda}\right)$$

for all (t, x) . This means that in the limit as $\lambda \downarrow 0$, we have

$$\|V^\pi - \widehat{V}^*\|_\infty = O\left(\lambda \log \frac{1}{\lambda}\right)$$

as required. □

2.3 Optimal Stopping Algorithm

Recall from the proof of Theorem 2.2 that $H(x, \pi, u) = (g(x) - u)\pi + \lambda R(\pi)$ is the Hamiltonian of the intensity-regularised problem. To solve the optimal stopping problem with reinforcement learning, we can use an iterative method that starts with $u_0 \in C^{1,2}$ arbitrarily chosen and alternating between:

- **Policy evaluation:** Solving the HJB equation

$$(\partial_t + \mathcal{L})u_n(t, x) + H(x, \pi_n(t, x), u_n(t, x)) = 0, \quad u_n(\cdot, T) = g \quad (29)$$

to find u_n .

- **Policy update:** Using u_n to find

$$\pi_{n+1}(t, x) = \arg \max_{\pi \geq 0} H(x, \pi, u_n(t, x)) = \exp\left(-\frac{u_n(t, x) - g(x)}{\lambda}\right) \quad (30)$$

As first shown in [7], this process ensures that as $n \rightarrow \infty$, (u_n) increases to V^* and π_n decreases to π^* .

Theorem 2.4 (Monotonicity of policy iteration). *Let $u_0 \in C_b(\mathbb{R})$ and define iteratively (u_n, π_n) using (29) and (30). Then the sequence of value functions satisfies*

$$u_0 \leq u_1 \leq \dots \leq u_n \leq V^*$$

for all $n \geq 0$. Moreover, the associated stopping intensities satisfy

$$\pi_1 \geq \pi_2 \geq \dots \geq \pi_n \geq \pi^*$$

pointwise. In particular,

$$u_n \uparrow V^* \quad \text{and} \quad \pi_n \downarrow \pi^*$$

pointwise as $n \rightarrow \infty$.

Proof. We divide the proof into three steps.

Step 1: Monotonicity of (u_n) . Fix $n \geq 0$ and define $\Delta_n = u_{n+1} - u_n$. Subtracting the linear PDEs satisfied by u_{n+1} and u_n yields

$$(\partial_t + \mathcal{L})\Delta_n + H(x, \pi_{n+1}, u_{n+1}) - H(x, \pi_n, u_n) = 0, \quad \Delta_n(\cdot, T) = 0.$$

Using $H(x, \pi, u) = (g(x) - u)\pi + \lambda R(\pi)$ and rearranging, we obtain

$$(\partial_t + \mathcal{L})\Delta_n - \pi_{n+1}\Delta_n = H(x, \pi_n, u_n) - H(x, \pi_{n+1}, u_n), \quad \Delta_n(\cdot, T) = 0.$$

By construction, $\pi_{n+1}(t, x)$ is a pointwise maximiser of $\pi \mapsto H(x, \pi, u_n(t, x))$, hence $H(x, \pi_n, u_n) - H(x, \pi_{n+1}, u_n) \leq 0$ pointwise. Since $\pi_{n+1} \geq 0$, the coefficient $-\pi_{n+1}$ is non-positive. Therefore, the parabolic maximum principle implies $\Delta_n \geq 0$, and hence

$$u_{n+1} \geq u_n$$

pointwise on $\mathbb{R} \times [0, T]$. Moreover, given that we have the relation $u_n = V^{\pi_n}$, and V^* satisfies

$$V^*(t, x) = \sup_{\pi \geq 0} V^\pi(t, x),$$

we immediately have

$$u_n(t, x) = V^{\pi_n}(t, x) \leq V^*(t, x)$$

for all (t, x) and $n \geq 0$. This gives us

$$u_0 \leq u_1 \leq \dots \leq u_n \leq V^*$$

for all $n \geq 0$.

Step 2: Monotonicity of (π_n) . From the policy update formula (30) and the fact that (u_n) is pointwise increasing, it follows that (π_n) is also pointwise decreasing, that is

$$\pi_1 \geq \pi_2 \geq \dots \geq \pi_n \geq \pi^*$$

on $\mathbb{R} \times [0, T]$ and for all $n \geq 1$.

Step 3: Convergence to (V^*, π^*) . Since (u_n) is pointwise increasing and bounded above by V^* , we have that the pointwise limit

$$u_\infty(t, x) = \lim_{n \rightarrow \infty} u_n(t, x)$$

exists. Moreover, the uniform bounds from Theorem 2.2 applied to the policies π_n yield a global bound

$$u_n(t, x) \leq \|g\|_\infty + \lambda(T - t), \quad 0 \leq t \leq T,$$

which is independent of n . Consequently, (π_n) is uniformly bounded above and below away from 0, which follows directly from the policy update (30). Similarly, for (π_n) we have

$$\pi_\infty(t, x) = \lim_{n \rightarrow \infty} \pi_{n+1}(t, x) = \lim_{n \rightarrow \infty} \exp\left(-\frac{u_n(t, x) - g(x)}{\lambda}\right) = \exp\left(-\frac{u_\infty(t, x) - g(x)}{\lambda}\right),$$

where the limit exists pointwise as (π_n) is pointwise decreasing and bounded from below by 0.

Now, note that for each n , u_{n+1} solves the linear terminal value problem

$$(\partial_t + \mathcal{L})u_{n+1} - \pi_{n+1}u_{n+1} = -g\pi_{n+1} - \lambda R(\pi_{n+1}), \quad u_{n+1}(T, \cdot) = g(\cdot),$$

with coefficients π_{n+1} uniformly bounded in L^∞ , by the bound on u_n and (30). Therefore, by the standard local parabolic a priori estimates (see [7] for more details), we have that $\{u_n\}$ is locally pre-compact, meaning along a sub-sequence $u_{n_k} \rightarrow u_\infty$ and the associated derivatives converge on compacts as well. This allows us to pass the limit to the evaluation equation for u_{n+1} , giving us

$$(\partial_t + \mathcal{L})u_\infty(t, x) + (g(x) - u_\infty(t, x))\pi_\infty(t, x) + \lambda R(\pi_\infty(t, x)) = 0, \quad u_\infty(T, \cdot) = g(\cdot)$$

pointwise on $\mathbb{R} \times [0, T]$. Finally, since $\pi_\infty(t, x)$ is the pointwise maximiser of $\pi \mapsto H(x, \pi, u_\infty(t, x))$, we have the identity

$$(g(x) - u_\infty(t, x))\pi_\infty(t, x) + \lambda R(\pi_\infty(t, x)) = \sup_{\pi \geq 0} H(x, \pi, u_\infty(t, x)) = \lambda \exp\left(-\frac{u_\infty(t, x) - g(x)}{\lambda}\right).$$

Thus u_∞ solves the optimal HJB equation

$$(\partial_t + \mathcal{L})u_\infty(t, x) + \lambda \exp\left(-\frac{u_\infty(t, x) - g(x)}{\lambda}\right) = 0, \quad u_\infty(T, \cdot) = g(\cdot).$$

By the uniqueness statement in Theorem 2.2 for the optimal HJB equation, we conclude that

$$u_\infty = V^*$$

pointwise. This further implies that

$$\pi_\infty(t, x) = \exp\left(-\frac{u_\infty(t, x) - g(x)}{\lambda}\right) = \exp\left(-\frac{V^*(t, x) - g(x)}{\lambda}\right) = \pi^*(t, x),$$

pointwise, completing the proof. $\square$

In a numerical setup, we approximate the value function using parameters θ^l for each time l to get $V_\theta(\cdot)$, since we cannot solve the HJB equation without knowing the full model. In our case, we will use a fully connected neural network. Regardless, the policy update admits a closed-form maximiser and Theorem 2.4 shows that each exact policy improvement step is monotone, meaning that we will converge to the optimal pair (V^*, π^*) as we increase the number of iterations N . Given a time grid $0 = t_0 < t_1 < \dots < t_L = T$, step size Δt and batch size M , the algorithm to solve the optimal stopping problem is as follows:

Algorithm 1 Entropy-regularised RL for optimal stopping via intensity control

- 1: **Input:** $\lambda > 0, r > 0$, learning rate η , iterations N
- 2: **Initialise:** parameters θ of $V_\theta(\cdot, \cdot)$; set $V_\theta(\cdot, T) \equiv g(\cdot)$
- 3: **for** $n = 1$ to N **do**
- 4: Generate M simulated paths $\{S_{t_l}^{(m)}\}_{l=0}^L$ of the environment
- 5: **for** $l = L - 1$ down to 0 **do**
- 6: **Policy update:** for each path m , set

$$\pi_{t_l}^{(m)} \leftarrow \exp\left(-\frac{V_\theta(S_{t_l}^{(m)}, t_l) - g(S_{t_l}^{(m)})}{\lambda}\right)$$

- 7: **TD residual:** for each m , define $R(\pi) = \pi - \pi \log \pi$ and

$$\delta_l^{(m)} = g(S_{t_l}^{(m)}) \pi_{t_l}^{(m)} \Delta t + \lambda R(\pi_{t_l}^{(m)}) \Delta t + e^{-r\Delta t} V_\theta(S_{t_{l+1}}^{(m)}, t_{l+1}) (1 - \pi_{t_l}^{(m)} \Delta t) - V_\theta(S_{t_l}^{(m)}, t_l)$$

- 8: **Regression step:** minimise $\sum_{m=1}^M (\delta_l^{(m)})^2$ by one gradient step

$$\theta \leftarrow \theta - \eta \nabla_\theta \sum_{m=1}^M (\delta_l^{(m)})^2$$

- 9: **Return:** trained V_θ and the intensity policy $\pi^*(S, t) = \exp(-[V_\theta(S, t) - g(S)]/\lambda)$
-

2.4 Numerical Example – American Put

We will implement and verify Algorithm 1 by applying it to a numerical example. Our focus here is the classical American put, where the payoff and underlying stock dynamics are

$$g(X) = (K - X)^+, \quad dX_t = X_t(r dt + \sigma dW_t).$$

In a financial context, the American put option is a contract that gives the holder the right to sell the underlying asset X to the issuer for the strike price K at any time before the expiration T . Intuitively, a put option contract becomes more valuable as the price of the underlying asset decreases, since the holder can sell for more than the current market price.

Using the reinforcement learning approach, the goal is to learn the value V derived from the model dX_t . Following the numerical benchmarks reported in [3], we will use the parameter set

$$X_0 = K = 40, \quad r = 0.06, \quad \sigma = 0.40, \quad T = 1.$$

In the algorithm, we discretise the time horizon into $L = 50$ points and use a separate neural network $V_\theta(\cdot, t_l)$ for each slice t_l to learn the model. Following [7], we parameterise the value as

$$V_\theta(S, t_l) = g(S) + f_\theta(S, g(S), t_l/T),$$

with f_θ represented as a fully connected network with width 21 and 2 hidden layers. This improves the stability of the learning process. Training is then performed on $M = 2^{10}$ realisations for $N = 3000$ iterations using the Adam optimiser with learning rate $\text{lr} = 10^{-3}$.

To evaluate the quality of the learned policies, we compute the value $\mathcal{P}$ as an out of sample estimate of the expected discounted payoff induced by the learned value function. Specifically, for each trained model we construct a stopping rule by comparing the learned value function with the immediate payoff. Along a simulated path $(X_{t_l})_{l=0}^L$, we stop at the first time index t_l such that $V_\theta(X_{t_l}, t_l) < g(X_{t_l})$, and otherwise exercise at maturity T . Mathematically, this corresponds to the boundary-type execution rule

$$\tau^\lambda = \inf\{t_l : V_\theta(X_{t_l}, t_l) \leq g(X_{t_l})\} \wedge T.$$

We use the pure stopping rule here to then compute the value (the fair price of the contract) $\mathcal{P}$ as the average

$$\mathcal{P} = \mathbb{E} \left[e^{-r\tau^\lambda} g(X_{\tau^\lambda}) \right],$$

where the expectation is approximated using an independent test set of 2^{18} simulated sample paths. This evaluation procedure reflects the performance of the learned randomised stopping policy when deployed as a deterministic execution boundary, and allows for a direct comparison with classical numerical benchmarks for the American put option.

To ensure the reliability of the results and minimise the chance error generated by the Monte Carlo sampling and stochastic gradient optimisation, we repeat the experiment 100 times. For each choice of λ , we aggregate the results over the 100 trials and finally report the mean final value and standard deviation. For comparison, we compare our results against $\mathcal{P}_{\text{ref}} = 5.311$ which was computed in [3] using a regression-based Monte Carlo method. While this value is itself obtained numerically, it provides a useful point of comparison for evaluating the convergence behaviour and residual bias from a reinforcement learning approach.

Figure 5 displays the error decay for $\lambda \in \{10, 0.1, 0.0001\}$, and table 1 summarises the final values obtained after $N = 3000$ training iterations. The behaviour is consistent with our previous theory:

- A larger λ accelerates the convergence but induces bias in the final result.
- A smaller λ reduces the bias but takes more iterations to converge to a solution.

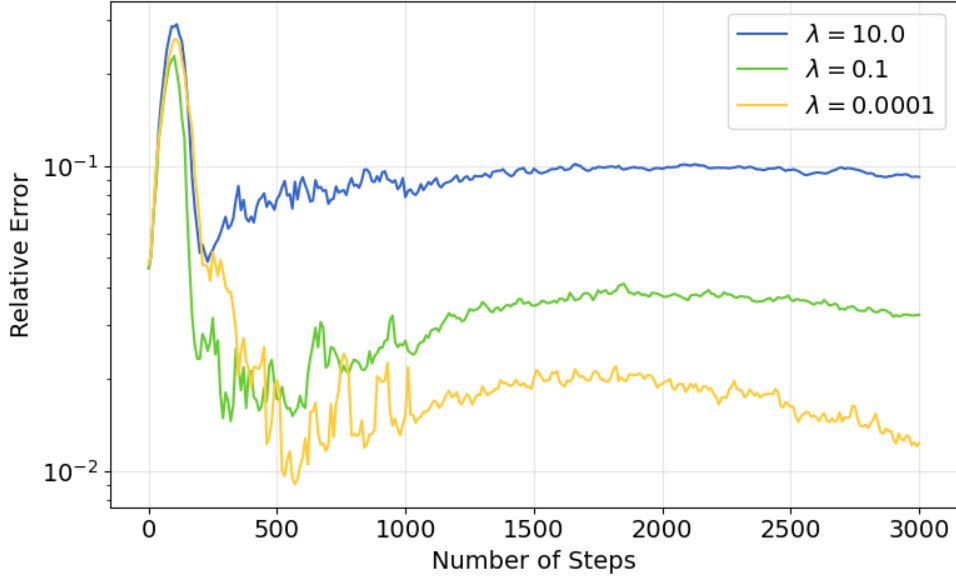


Figure 5: Learning curves for three choices of λ .

λ	Value $\mathcal{P}$	SD	Relative error
10	4.8224	0.054	0.092
0.1	5.1384	0.037	0.033
0.0001	5.2456	0.031	0.012

Table 1: Final performance of Algorithm 1 for different values of λ .

3 Double Stopping Problems

Although the theory behind the single optimal stopping problem is well-researched, there are several applications in practice that require multiple stopping opportunities and therefore cannot be solved using this framework. For this section, we work with a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where once again $\mathcal{T}_{[0,T]}$ denotes the set of $(\mathcal{F}_t)$ -stopping times taking values in $[0, T]$. We will also work in the finite time horizon case ($T < \infty$), however the results here can be extended to the infinite time horizon case.

We now consider the optimal double stopping problem, where the agent instead selects two stopping times (τ_1, τ_2) to optimise a joint payoff function $\psi(\tau_1, \tau_2)$. Double stopping and multiple stopping problems often appear in finance, particularly in exotic options. One example is an exchange option: given two risky assets X^1 and X^2 , the holder chooses (τ_1, τ_2) and at time $\tau_1 \vee \tau_2$ receives the payoff

$$\psi(\tau_1, \tau_2) = e^{-r(\tau_1 \vee \tau_2)} (X_{\tau_1}^1 - X_{\tau_2}^2)^+.$$

This problem was originally defined in [19], and adapted to double stopping in [16]. For this problem, we will need to extend the concept of admissible policies in section 2 to the double stopping setup.

Definition 3.1. We say that the family

$$\{\psi(\theta, S), \theta, S \in \mathcal{T}_{[0,T]}\}$$

is *biadmissible* if for all $\theta, S \in \mathcal{T}_{[0,T]}$, $\psi(\theta, S)$ is a $\mathcal{F}_{\theta \vee S}$ -measurable positive random variable and for all $\theta, \theta', S, S' \in \mathcal{T}_{[0,T]}$, $\psi(\theta, S) = \psi(\theta', S')$ almost surely on $\{\theta = \theta'\} \cap \{S = S'\}$.

Note that we work at the level of random variables indexed by stopping times since in general a two-time payoff $\psi(\tau_1, \tau_2)$ does not aggregate into a single adapted reward process evaluated at one time, whereas the Snell envelope and essential supremum techniques from before extend naturally to biadmissible families. We also need to update our definition of the value function.

Definition 3.2. The *value function* of the double stopping problem from time S is defined as:

$$v(S) = \operatorname{ess\,sup}_{\tau_1, \tau_2 \in \mathcal{T}_{[S,T]}} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S].$$

This framework has several key applications across industries, since multiple stopping problems appear when a decision maker needs to select several stopping times, and the payoff depends on the times chosen and their ordering [26]. In finance, any trading strategy requires you to select an entry and exit point to execute the trade, naturally leading to a double stopping setup. In addition, several exotic derivative products, such as exchange or swing options [4], require the holder to select multiple exercise times to achieve an optimal payoff.

3.1 Reduction to Single Stopping

A key theoretical result in [16] shows that the double stopping problem can be reduced to a sequence of single stopping problems. Although the payoff $\psi(\tau_1, \tau_2)$ depends on two stopping times and therefore does not generally aggregate into a single adapted reward process, the problem can still be decomposed by conditioning on the first exercise time. Indeed, if we set $\theta = \tau_1 \wedge \tau_2$ as the first time at which either right is exercised, then at time θ the agent now has a one-step problem: either τ_2 has occurred and they need to optimise their choice of τ_1 ($u_1(\theta)$), or τ_1 has occurred and they need to optimise for τ_2 ($u_2(\theta)$). Taking $\phi(\theta) = \max\{u_1(\theta), u_2(\theta)\}$ therefore encodes the best continuation value available at the first exercise time, and the original double stopping optimisation reduces to a single optimal stopping problem with reward $\phi(\cdot)$. From a computational perspective, this decomposition is particularly valuable as it splits the double stopping problem into three single stopping problems, which can each be solved following Algorithm 1.

We will make the proof of the reduction principle more rigorous here by working in the more general case where there may not exist solutions to the single stopping problems.

Theorem 3.1 (Reduction Principle). *Let $\{\psi(\theta, S), \theta, S \in \mathcal{T}_{[0, T]}\}$ be a biadmissible family of non-negative $\mathcal{F}_{\theta \vee S}$ -measurable random variables. For each $\theta \in \mathcal{T}_{[0, T]}$, define the single-stopping value functions*

$$u_1(\theta) = \operatorname{ess\,sup}_{\tau_1 \in \mathcal{T}_{[\theta, T]}} \mathbb{E}[\psi(\tau_1, \theta) \mid \mathcal{F}_\theta], \quad u_2(\theta) = \operatorname{ess\,sup}_{\tau_2 \in \mathcal{T}_{[\theta, T]}} \mathbb{E}[\psi(\theta, \tau_2) \mid \mathcal{F}_\theta],$$

and the reduced reward

$$\phi(\theta) = \max(u_1(\theta), u_2(\theta)),$$

with its corresponding one-stopping value

$$u(S) = \operatorname{ess\,sup}_{\theta \in \mathcal{T}_{[S, T]}} \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S].$$

Then for every stopping time $S \in \mathcal{T}_{[0, T]}$, we have

$$v(S) = u(S)$$

almost surely.

Proof. Fix a stopping time S . We will split this proof into two steps.

Step 1: $v(S) \leq u(S)$. Fix $\tau_1, \tau_2 \in \mathcal{T}_{[S, T]}$ and define the event $A = \{\tau_1 \leq \tau_2\} \in \mathcal{F}_{\tau_1 \vee \tau_2}$. If we split $\psi(\tau_1, \tau_2)$ on A , we get

$$\mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S] = \mathbb{E}[1_A \psi(\tau_1, \tau_2) + 1_{A^c} \psi(\tau_1, \tau_2) \mid \mathcal{F}_S].$$

On A , we have $\mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_1}] \leq u_2(\tau_1)$ and hence

$$1_A \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_1 \vee \tau_2}] = 1_A \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_1}] \leq 1_A u_2(\tau_1) \leq 1_A \phi(\tau_1 \wedge \tau_2).$$

Similarly, on A^c we have $\mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_2}] \leq u_1(\tau_2)$ and thus

$$1_{A^c} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_1 \vee \tau_2}] = 1_{A^c} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_{\tau_2}] \leq 1_{A^c} u_1(\tau_2) \leq 1_{A^c} \phi(\tau_1 \wedge \tau_2).$$

Putting these together yields

$$\mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S] \leq \mathbb{E}[\phi(\tau_1 \wedge \tau_2) \mid \mathcal{F}_S].$$

Since $\tau_1 \wedge \tau_2 \in \mathcal{T}_{[S, T]}$, taking the essential supremum over all τ_1, τ_2 gives

$$v(S) = \operatorname{ess\,sup}_{\tau_1, \tau_2 \in \mathcal{T}_{[S, T]}} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S] \leq \operatorname{ess\,sup}_{\tau_1, \tau_2 \in \mathcal{T}_{[S, T]}} \mathbb{E}[\phi(\tau_1 \wedge \tau_2) \mid \mathcal{F}_S] = u(S)$$

almost surely, where for the last equality we have used the fact that $\{\tau_1 \wedge \tau_2 : \tau_1, \tau_2 \in \mathcal{T}_{[S, T]}\} = \mathcal{T}_{[S, T]}$. This makes sense, since for any $\theta \in \mathcal{T}_{[S, T]}$ we may take $\tau_1 = \theta$ and $\tau_2 = T$ to get $\tau_1 \wedge \tau_2 = \theta$.

Step 2: $v(S) \geq u(S)$. For the reverse inequality we will follow a standard approximation argument. To do so, we will construct an approximately optimal $\theta \in \mathcal{T}_{[S, T]}$ for $u(S)$, giving us an approximately optimal pair (τ_1, τ_2) for the double stopping problem.

Fix $\varepsilon > 0$. Now, following the definition of $u(S)$, choose $\theta \in \mathcal{T}_{[S, T]}$ such that

$$\mathbb{E}[\phi(\theta) \mid \mathcal{F}_S] \geq u(S) - \varepsilon.$$

This means that on the event $B = \{u_1(\theta) \geq u_2(\theta)\}$, we may further pick $\theta_1 \in \mathcal{T}_{[\theta, T]}$ to get

$$1_B \mathbb{E}[\psi(\theta_1, \theta) \mid \mathcal{F}_\theta] \geq 1_B (u_1(\theta) - \varepsilon).$$

Similarly, on $B^c = \{u_1(\theta) < u_2(\theta)\}$ we can choose $\theta_2 \in \mathcal{T}_{[\theta, T]}$ to get

$$1_{B^c} \mathbb{E}[\psi(\theta, \theta_2) \mid \mathcal{F}_\theta] \geq 1_{B^c} (u_2(\theta) - \varepsilon).$$

Define the pair of stopping times

$$\tau_1 = \theta_1 1_B + \theta 1_{B^c}, \quad \tau_2 = \theta 1_B + \theta_2 1_{B^c}.$$

Since the selection event $B = \{u_1(\theta) \geq u_2(\theta)\}$ is $\mathcal{F}_\theta$ -measurable, τ_1 and τ_2 must belong in $\mathcal{T}_{[S, T]}$. Thus, by the tower property we obtain

$$\begin{aligned} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S] &= \mathbb{E}[\mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_\theta] \mid \mathcal{F}_S] \\ &= \mathbb{E}[\mathbb{E}[1_B \psi(\theta_1, \theta) + 1_{B^c} \psi(\theta, \theta_2) \mid \mathcal{F}_\theta] \mid \mathcal{F}_S] \\ &= \mathbb{E}[1_B \mathbb{E}[\psi(\theta_1, \theta) \mid \mathcal{F}_\theta] + 1_{B^c} \mathbb{E}[\psi(\theta, \theta_2) \mid \mathcal{F}_\theta] \mid \mathcal{F}_S] \\ &\geq \mathbb{E}[1_B (u_1(\theta) - \varepsilon) + 1_{B^c} (u_2(\theta) - \varepsilon) \mid \mathcal{F}_S] \\ &= \mathbb{E}[\max\{u_1(\theta), u_2(\theta)\} - \varepsilon \mid \mathcal{F}_S] \\ &= \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S] - \varepsilon. \end{aligned}$$

Therefore, after taking the essential supremum we get

$$v(S) = \operatorname{ess\,sup}_{\tau_1, \tau_2} \mathbb{E}[\psi(\tau_1, \tau_2) \mid \mathcal{F}_S] \geq \operatorname{ess\,sup}_{\theta} \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S] = u(S) - \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, we conclude $v(S) \geq u(S)$ almost surely.

Combining Steps 1 and 2 gives us $v(S) = u(S)$ almost surely. This proves the reduction principle. $\square$

As we can see from Theorem 3.1, the original double stopping problem is equivalent to a single stopping problem with reward ϕ :

$$v(S) = \operatorname{ess\,sup}_{\theta \in \mathcal{T}_{[S, T]}} \mathbb{E}[\phi(\theta) \mid \mathcal{F}_S].$$

Once an optimal θ^* for $\phi(\theta^*) = \max(u_1(\theta^*), u_2(\theta^*))$ has been found, the corresponding pair (τ_1^*, τ_2^*) can then be determined by solving the two single stopping sub-problems

$$u_1(\theta^*) = \operatorname{ess\,sup}_{\tau_1 \in \mathcal{T}_{[\theta^*, T]}} \mathbb{E}[\psi(\tau_1, \theta^*) \mid \mathcal{F}_{\theta^*}], \quad u_2(\theta^*) = \operatorname{ess\,sup}_{\tau_2 \in \mathcal{T}_{[\theta^*, T]}} \mathbb{E}[\psi(\theta^*, \tau_2) \mid \mathcal{F}_{\theta^*}].$$

It has been shown in [16] that if the biadmissible reward family $\{\psi(\theta, S), \theta, S \in \mathcal{T}_{[0, T]}\}$ is uniformly continuous in expectation (UCE) along stopping times, that is

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\operatorname{ess\,sup}_{\theta \in \mathcal{T}_{[0, T]}} |\psi(\theta, S) - \psi(\theta, S_n)| \right] = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \mathbb{E} \left[\operatorname{ess\,sup}_{\theta \in \mathcal{T}_{[0, T]}} |\psi(S, \theta) - \psi(S_n, \theta)| \right] = 0,$$

with $S_n \rightarrow S$, and satisfies the L^p bound $\mathbb{E}[\operatorname{ess\,sup}_{\theta, S \in \mathcal{T}_{[0, T]}} \psi(\theta, S)^p] < \infty$ for some $p > 1$, then the reduced family $\{\phi(\theta), \theta \in \mathcal{T}_{[0, T]}\}$ is also continuous in expectation. This ensures that there exists an optimal θ^* in the associated stopping problem. In our case, this means that all three single stopping problems can be solved using the reinforcement learning framework developed in section 2. This approach can further generalise to multiple stopping problems by iteratively applying the same reduction principle. See also [26] for a game-theoretic double stopping setting, and [11] for geometric Brownian motion (GBM) maxima and minima applications.

3.2 Double Stopping Algorithm

Here we present an algorithm to numerically solve the double stopping problem with reinforcement learning. Following the theory in the previous section, we will decompose the problem into multiple single stopping problems to solve separately, following a divide-and-conquer approach.

Algorithm 2 Entropy-regularised RL for double stopping via reduction

- 1: **Input:** $\lambda > 0$, $\alpha > 0$, learning rates $\eta_{\text{in}}, \eta_{\text{out}}$, inner/outer iterations $N_{\text{in}}, N_{\text{out}}$
- 2: **Setup:** Time grid $0 = t_0 < \dots < t_L = T$, step size Δt , batch size M
- 3: **Initialise:** Inner values $u_1(\cdot, t_l | \theta)$ and $u_2(\cdot, t_l | \theta)$, outer value $v(\cdot, t_l)$ for each slice t_l .
- 4: Terminal conditions $u_1(\cdot, T | \theta) \equiv \psi(T, \theta)$, $u_2(\cdot, T | \theta) \equiv \psi(\theta, T)$, $v(\cdot, T) \equiv \psi(T, T)$.

Stage A: Train the inner problems u_1, u_2 .

- 5: **for** $n = 1$ to N_{in} **do**
- 6: Generate M simulated paths $\{S_{t_l}^{(m)}\}_{l=0}^L$ of the environment
- 7: **for** $k = 0$ to L **do** $\triangleright \theta = t_k$
- 8: **for** $l = L - 1$ down to k **do**
- 9: **Inner-1 policy update at $(t_l | \theta = t_k)$:** for each m , set

$$\pi_{t_l|t_k}^{1,(m)} \leftarrow \exp\left(-\frac{u_1(S_{t_l}^{(m)}, t_l|t_k) - \psi^{(m)}(t_l, t_k)}{\lambda}\right)$$

- 10: **Inner-1 TD residual:** with $R(\pi) = \pi - \pi \log \pi$, define for each m

$$\delta_{l|k}^{1,(m)} = \psi^{(m)}(t_l, t_k) \pi_{t_l|t_k}^{1,(m)} \Delta t + \lambda R(\pi_{t_l|t_k}^{1,(m)}) \Delta t + e^{-r\Delta t} u_1(S_{t_{l+1}}^{(m)}, t_{l+1} | t_k) (1 - \pi_{t_l|t_k}^{1,(m)} \Delta t) - u_1(S_{t_l}^{(m)}, t_l | t_k)$$

- 11: **Inner-1 regression:** one gradient step on $\sum_{m=1}^M (\delta_{l|k}^{1,(m)})^2$ with step η_{in}
- 12: **Inner-2 policy update at $(t_l | \theta = t_k)$:** for each m , set

$$\pi_{t_l|t_k}^{2,(m)} \leftarrow \exp\left(-\frac{u_2(S_{t_l}^{(m)}, t_l|t_k) - \psi^{(m)}(t_k, t_l)}{\lambda}\right)$$

- 13: **Inner-2 TD residual:** for each m

$$\delta_{l|k}^{2,(m)} = \psi^{(m)}(t_k, t_l) \pi_{t_l|t_k}^{2,(m)} \Delta t + \lambda R(\pi_{t_l|t_k}^{2,(m)}) \Delta t + e^{-r\Delta t} u_2(S_{t_{l+1}}^{(m)}, t_{l+1} | t_k) (1 - \pi_{t_l|t_k}^{2,(m)} \Delta t) - u_2(S_{t_l}^{(m)}, t_l | t_k)$$

- 14: **Inner-2 regression:** one gradient step on $\sum_{m=1}^M (\delta_{l|k}^{2,(m)})^2$ with step η_{in}
-

Stage B: Train the outer problem v .

 15: **for** $n = 1$ to N_{out} **do**

 16: Generate M simulated paths $\{S_{t_l}^{(m)}\}_{l=0}^L$

 17: **for** $l = L - 1$ down to 0 **do**

 18: **Outer reward at t_l :** for each m , set

$$\phi_{t_l}^{(m)} \leftarrow \max \left\{ u_1(S_{t_l}^{(m)}, t_l \mid t_l), u_2(S_{t_l}^{(m)}, t_l \mid t_l) \right\}$$

 19: **Outer policy update:** for each m , set

$$\varrho_{t_l}^{(m)} \leftarrow \exp \left(- \frac{v(S_{t_l}^{(m)}, t_l) - \phi_{t_l}^{(m)}}{\lambda} \right)$$

 20: **Outer TD residual:** for each m

$$\delta_l^{(m)} = \phi_{t_l}^{(m)} \varrho_{t_l}^{(m)} \Delta t + \lambda R(\varrho_{t_l}^{(m)}) \Delta t + e^{-r\Delta t} v(S_{t_{l+1}}^{(m)}, t_{l+1}) (1 - \varrho_{t_l}^{(m)} \Delta t) - v(S_{t_l}^{(m)}, t_l)$$

 21: **Outer regression:** one gradient step on $\sum_{m=1}^M (\delta_l^{(m)})^2$ with step η_{out}

 22: **Return:** trained u_1, u_2, v and the intensity policies

$$\pi_1^*(S, t \mid \theta) = \exp(-[u_1(S, t \mid \theta) - \psi(t, \theta)]/\lambda), \quad \pi_2^*(S, t \mid \theta) = \exp(-[u_2(S, t \mid \theta) - \psi(\theta, t)]/\lambda),$$

$$\varrho^*(S, t) = \exp(-[v(S, t) - \max\{u_1(S, t \mid t), u_2(S, t \mid t)\}]/\lambda).$$

 23: **Construct unordered (τ_1^*, τ_2^*) :** simulate ϱ^* forward; upon outer stop at $\hat{\theta}$, compare $u_1(\cdot, \hat{\theta} \mid \hat{\theta})$ and $u_2(\cdot, \hat{\theta} \mid \hat{\theta})$; if $u_1 \geq u_2$, set $\tau_2^* = \hat{\theta}$ and sample τ_1^* using $\pi_1^*(\cdot \mid \hat{\theta})$; else set $\tau_1^* = \hat{\theta}$ and sample τ_2^* using $\pi_2^*(\cdot \mid \hat{\theta})$.

By Theorem 2.4 and Theorem 3.1, this algorithm will converge to the optimal solution as $N_{\text{in}} \rightarrow \infty$ and $N_{\text{out}} \rightarrow \infty$. Moreover, we can extend this methodology to larger problems with more stopping times following the same reduction principle. It should be noted that for any n -stopping problem, the reduction principle splits it into $2^n - 1$ unique single stopping problems, leading to an exponential runtime as n increases.

3.3 Numerical Example – Exchange Option

We now demonstrate the reduction principle from subsection 3.1 on a concrete example and solve a double stopping problem by solving two single stopping sub-problems in sequence. Specifically, we consider the exchange option introduced earlier, which has the biadmissible reward family

$$\psi(\tau_1, \tau_2) = e^{-r(\tau_1 \vee \tau_2)} (X_{\tau_1}^1 - X_{\tau_2}^2)^+,$$

which has a theoretical value (price) of

$$\mathcal{P} = \sup_{\tau_1, \tau_2 \in \mathcal{T}_{[0, T]}} \mathbb{E} \left[e^{-r(\tau_1 \vee \tau_2)} (X_{\tau_1}^1 - X_{\tau_2}^2)^+ \right].$$

We model (X^1, X^2) under the risk-neutral measure as correlated GBM paths

$$dX_t^i = r X_t^i dt + \sigma X_t^i dW_t^i, \quad i = 1, 2, \quad d\langle W^1, W^2 \rangle_t = \rho dt.$$

We first check that an optimal solution exists by verifying the uniformly continuous in expectation assumption. Fix $S \in \mathcal{T}_{[0, T]}$ and let $S_n \downarrow S$ almost surely. Using the Lipschitz property of the positive part,

$$|(a - b)^+ - (a - c)^+| \leq |b - c|,$$

we have for every $\theta \in \mathcal{T}_{[0, T]}$,

$$\begin{aligned} |\psi(\theta, S) - \psi(\theta, S_n)| &= \left| e^{-r(\theta \vee S)} (X_\theta^1 - X_S^2)^+ - e^{-r(\theta \vee S_n)} (X_\theta^1 - X_{S_n}^2)^+ \right| \\ &\leq \left| (X_\theta^1 - X_S^2)^+ - (X_\theta^1 - X_{S_n}^2)^+ \right| + \left| e^{-r(\theta \vee S)} - e^{-r(\theta \vee S_n)} \right| (X_\theta^1 + X_S^2 + X_{S_n}^2) \\ &\leq |X_S^2 - X_{S_n}^2| + \left| e^{-r(\theta \vee S)} - e^{-r(\theta \vee S_n)} \right| (X_\theta^1 + X_S^2 + X_{S_n}^2). \end{aligned}$$

Taking $\text{ess sup}_{\theta \in \mathcal{T}_{[0, T]}}$ yields

$$\text{ess sup}_{\theta \in \mathcal{T}_{[0, T]}} |\psi(\theta, S) - \psi(\theta, S_n)| \leq |X_S^2 - X_{S_n}^2| + \text{ess sup}_{\theta \in \mathcal{T}_{[0, T]}} \left| e^{-r(\theta \vee S)} - e^{-r(\theta \vee S_n)} \right| (X_\theta^1 + X_S^2 + X_{S_n}^2).$$

By continuity of X^2 , we have $X_{S_n}^2 \rightarrow X_S^2$ almost surely. Moreover, since e^{-rt} is Lipschitz on $[0, T]$ and $\theta \vee S_n \rightarrow \theta \vee S$ almost surely, the discount term converges to 0 almost surely. Moreover, the required L^p integrability holds in this model. Indeed, since $e^{-r(\theta \vee S)} \leq 1$ and $(x - y)^+ \leq x$ for all $x, y \geq 0$, we have

$$0 \leq \psi(\theta, S) \leq X_\theta^1 \leq \sup_{0 \leq t \leq T} X_t^1,$$

and hence for any $p > 1$,

$$\text{ess sup}_{\theta, S \in \mathcal{T}_{[0, T]}} \psi(\theta, S)^p \leq \left(\sup_{0 \leq t \leq T} X_t^1 \right)^p.$$

Since X^1 follows geometric Brownian motion, $(X_t^1)^p$ is a non-negative submartingale for any $p > 1$ we can use Doob's L^p maximal inequality to obtain

$$\begin{aligned} \mathbb{E} \left[\sup_{0 \leq t \leq T} (X_t^1)^p \right] &\leq \left(\frac{p}{p-1} \right)^p \mathbb{E}[(X_T^1)^p] \\ &= \left(\frac{p}{p-1} \right)^p (X_0^1)^p \exp \left(prT + \frac{1}{2} p(p-1) \sigma^2 T \right) \\ &< \infty, \end{aligned}$$

and therefore

$$\mathbb{E} \left[\text{ess sup}_{\theta, S \in \mathcal{T}_{[0, T]}} \psi(\theta, S)^p \right] < \infty$$

for any $p > 1$. Consequently, using the almost sure convergence established above and dominated convergence, we obtain

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\text{ess sup}_{\theta \in \mathcal{T}_{[0, T]}} |\psi(\theta, S) - \psi(\theta, S_n)| \right] = 0$$

3.3 Numerical Example – Exchange Option

as required. The second UCE condition, $\lim_{n \rightarrow \infty} \mathbb{E}[\text{ess sup}_{\theta} |\psi(S, \theta) - \psi(S_n, \theta)|] = 0$, follows similarly by swapping the arguments. Therefore the UCE assumption in [16] is satisfied in this setting.

For this experiment, we use the parameters

$$X_0^1 = X_0^2 = 60, \quad r = 0.06, \quad \sigma = 0.40, \quad \rho = 0.50, \quad T = 1.$$

The horizon is discretised on a Bermudan grid of $L = 50$ time points ($\Delta t = T/L$), and the two-dimensional geometric Brownian motion is simulated pathwise at each grid step. Similarly to subsection 2.4, at each time slice t_l we parameterise the value by the additive ansatz

$$V_{\theta}(\text{state}, t_l) = g(\text{state}) + f_{\theta}(\text{features}, g(\text{state}), t_l/T),$$

and use the randomised stopping intensity (28) when updating the policy. We investigate the impact of the exploration parameter by setting $\lambda \in \{10, 0.1, 0.0001\}$.

To approximate f , we train ReLU neural networks with 2 hidden layers and a width of 21 per slice using the Adam optimiser with learning rate $\text{lr} = 10^{-3}$. We perform the training in batch sizes of $M = 2^{10}$ and use an independent test set of size 2^{18} for evaluation. Following the reduction in subsection 3.1, we solve the exchange option by training three single stopping problems: two inner problems to compute the reduced reward $\phi(\theta) = \max\{u_1(\theta), u_2(\theta)\}$, and then an outer stopping problem for ϕ .

The final computed values of the exchange option for each λ are shown in Table 2. Figure 6 displays the learning curves for the inner and outer stages for each λ . On each path we implement the Bermudan stopping rule to stop at the first grid time l such that $V_l \leq g_l$, and report the Monte Carlo estimate of the mean discounted payoff. Once again, we see that a larger λ leads to faster and smoother convergence, while a smaller λ reduces the bias at the cost of slower learning.

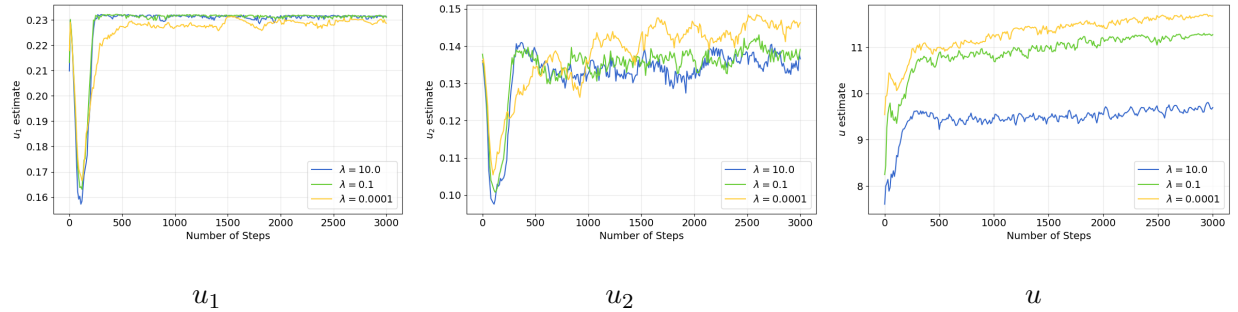


Figure 6: Learning curves for the exchange option.

λ	Value $\mathcal{P}$	SD
10	9.6966	0.356
0.1	11.2723	0.302
0.0001	11.6747	0.370

Table 2: Final performance of Algorithm 2 for different values of λ .

4 Time-Inconsistent Stopping – Discrete-Time Case

In previous optimal stopping problems we have been using exponential discounting, making preferences time-consistent. If we instead use non-exponential discounting, the preferences of one's future self may disagree with those of the present, making the problem time-inconsistent. We will follow the same setup as [14]. Let X_t be a discrete Markov process on a Polish space $\mathbb{X}$ with transition kernel Q . Specifically, for any $x \in \mathbb{X}$, $A \in \mathcal{B}(\mathbb{X})$ and all $t = 0, 1, \dots$, we have:

$$P(X_{t+1} \in A \mid X_t = x) = \int_A Q(x, dy).$$

Definition 4.1. The *time-inconsistent stopping objective function* is defined as

$$V^\tau(x) = \mathbb{E}_x [\delta(\tau)g(X_\tau)]$$

where $g \geq 0$ is a bounded *payoff function* and $\delta : [0, \infty) \rightarrow [0, 1]$ is a strictly decreasing *discount function* satisfying:

- $\delta(0) = 1$.
- $\lim_{k \rightarrow \infty} \delta(k) = 0$.
- $\delta(i)\delta(j) \leq \delta(i+j)$ for all $i, j \in \mathbb{N} \cup \{0\}$.

The last assumption is known as the decreasing impatience (DI) property, and is crucial when working in the time-inconsistent setting. It is related to the idea of decreasing impatience in behavioural economics developed in [28]. Intuitively, when choosing between two rewards that are further away in time, people are more willing to wait to attain the larger reward.

Once again, the goal is to find an optimal stopping time $\tau^*(x)$ that maximises the expected discounted payoff

$$V^*(x) = \sup_{\tau \in \mathcal{T}} V^\tau(x),$$

where $\mathcal{T}$ is the set of admissible stopping times. We say that the problem is *time-consistent* if the optimal stopping time from time 0 is the same as the optimal stopping time from $t > 0$, provided $\tau^* \geq t$. In particular, when δ is no longer exponential, this may no longer be the case, leading to a *time-inconsistent* problem. Instead of a single stopping time, we now work with a *stopping policy* $\tau : \mathbb{X} \rightarrow \{0, 1\}$, which specifies the agent to stop at the first time s when $\tau(X_s) = 0$. This can be equivalently formulated as a stopping region $\tau(x) = 1_{S^c}(x)$ where $S = \{x \in \mathbb{X} : \tau(x) = 0\}$. We think of the problem from a game-theoretic perspective: given that the agent will use S as the stopping region, we choose the optimal stopping strategy for today in response. At each time point, the agent once again has two options:

- Stop, and receive $g(x)$ immediately.
- Continue, and eventually stop at $\rho(x, S) = \inf\{t \geq 1 : X_t \in S\} \in \mathcal{T}$, yielding the payoff $V^{\rho(x, S)}(x)$.

This sets the optimal stopping strategy for today to be $\Theta(S) = \{x \in \mathbb{X} : g(x) \geq V^{\rho(x,S)}(x)\}$ [14]. Intuitively, given a strategy S , $\Theta(S)$ represents the optimal strategy to use today if we believe our future self will follow S . This means that our original goal of solving the optimal stopping problem shifts to finding an equilibrium.

Definition 4.2. We say that S is an *equilibrium stopping region* (denoted $S \in \mathcal{E}$) if we have a fixed point $\Theta(S) = S$, where $\mathcal{E}$ is the set of all equilibria.

This is equivalent to:

$$\begin{cases} g(x) \geq V^{\rho(x,S)}(x), & x \in S \\ g(x) < V^{\rho(x,S)}(x), & x \in \mathbb{X} \setminus S \end{cases}$$

To find an equilibrium, we can use a fixed-point iteration method to find $S_\infty = \lim_{n \rightarrow \infty} \Theta^n(S)$ [14].

4.1 Application to Reinforcement Learning

We will now apply this theory of time-inconsistent optimal stopping to our reinforcement learning framework. Instead of a hard stopping region S , we use the *mapping* $\pi : \mathbb{X} \rightarrow [0, 1]$ to determine the probability of stopping at any point.

Definition 4.3. Given a policy π , we define the randomised (relaxed) stopping time as

$$\tau_\pi = \inf\{k \geq 0 : U_k \leq \pi(X_k)\},$$

where $(U_k)_{k \geq 0}$ are independent and identically distributed (iid) $U(0, 1)$ random variables, all of which are independent of X .

To encourage exploration, we incorporate Shannon's differential entropy

$$H(\pi(x)) = -\pi(x) \log \pi(x) - (1 - \pi(x)) \log(1 - \pi(x)) \quad (31)$$

into the objective function

$$V^\pi(x) = \mathbb{E}_x \left(\delta(\tau)g(X_\tau) + \lambda \sum_{k=0}^{\tau} \delta(k)H(\pi(X_k)) \right), \quad (32)$$

as well as the future value after continuing for one step

$$J^\pi(x) = \mathbb{E}_x \left(\delta(\tau)g(X_\tau) + \lambda \sum_{k=1}^{\tau} \delta(k)H(\pi(X_k)) \mid \tau \geq 1 \right), \quad (33)$$

where $\lambda > 0$ is the exploration parameter. Similarly, we use a game-theoretic perspective to guide our decision-making.

Given the current underlying state x and the agent's future plan to follow the policy $\tilde{\pi}(x)$, we either:

- Stop with probability $\pi(x)$, and receive $g(x)$.
- Continue with probability $1 - \pi(x)$, and follow the future policy $\tilde{\pi}$ to receive (in expectation) $J^{\tilde{\pi}}(x)$.

The above logic yields the *entropy-regularised one-step* objective function

$$U_\lambda(x, \pi, \tilde{\pi}) = \pi(x)g(x) + (1 - \pi(x))J^{\tilde{\pi}}(x) + \lambda H(\pi(x)) \quad (34)$$

which, when maximised, determines the current optimal choice of $\pi(x)$.

From this definition, we also have that $V^\pi(x) = U_\lambda(x, \pi, \pi)$ and

$$\begin{aligned} J^\pi(x) &= \mathbb{E}_x \left[\sum_{k=1}^{\infty} \delta(k) \left(\prod_{j=1}^{k-1} (1 - \pi(X_j)) \right) V^\pi(X_k) \right] \\ &= \mathbb{E}_x \left[\sum_{k=1}^{\infty} \delta(k) \left(\prod_{j=1}^{k-1} (1 - \pi(X_j)) \right) U_\lambda(X_k, \pi, \pi) \right]. \end{aligned}$$

Given that U_λ is concave in π , the unique maximiser for $\pi(x)$ in (34) is:

$$\pi^*(x) = \arg \max_{\pi} U_\lambda(x, \pi, \tilde{\pi}) = \frac{1}{1 + \exp\left(\frac{J^{\tilde{\pi}}(x) - g(x)}{\lambda}\right)} \in (0, 1). \quad (35)$$

A plot of the unique maximiser for varying λ is shown in Figure 7. In line with the deterministic setup, we will define an equilibrium as the point when the future policy coincides with the current optimal policy.

Definition 4.4. We say that π^* is an equilibrium if

$$\pi^*(x) \in \arg \max_{\pi} U_\lambda(x, \pi, \pi^*)$$

for all $x \in \mathbb{X}$.

To find an equilibrium, we follow an iterative process: given an initial policy π_0 , we determine π_1 using (35) with $\tilde{\pi} = \pi_0$ set as the future policy. We then repeat this process until convergence to obtain the equilibrium $\pi_\infty(x) = \lim_{n \rightarrow \infty} \pi_n(x)$. For equilibria and time consistency under alternative risk criteria, see [2].

We now establish the existence of an equilibrium more rigorously by adapting the proofs of Lemma 3.1, Theorem 3.1 and Corollary 3.1 respectively from [14]. We emphasise that δ must satisfy the decreasing impatience property for the claims to hold.

Lemma 4.1 (Monotonicity). *Suppose $\pi_1(x) = \arg \max_{\pi} U_{\lambda}(x, \pi, \pi_0) \leq \pi_0(x)$ for all $x \in \mathbb{X}$. Then for all $x \in \mathbb{X}$,*

$$J^{\pi_0}(x) \leq J^{\pi_1}(x).$$

In particular, $\pi_2(x) = \arg \max_{\pi} U_{\lambda}(x, \pi, \pi_1) \leq \pi_1(x)$ for all $x \in \mathbb{X}$.

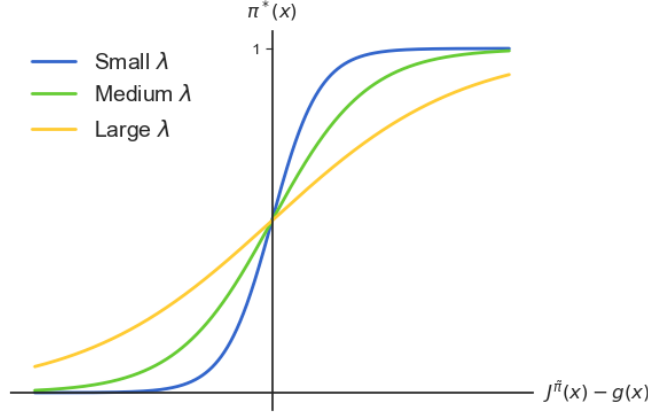


Figure 7: Plot of π^* for varying λ .

Proof. Fix $X_0 = x \in \mathbb{X}$ and $\pi_1 \leq \pi_0$. We work on a common probability space carrying X and an iid sequence $(U_k)_{k \geq 1}$ with $U_k \sim \text{Unif}(0, 1)$, independent of X . Under this coupling, $\pi_1 \leq \pi_0$ implies the corresponding stopping times satisfy

$$\tau_{\pi_1} \geq \tau_{\pi_0}$$

almost surely. For $m \geq 0$, define the hybrid policy

$$\Pi^{(m)}(X_k) = \begin{cases} \pi_1(X_k), & k \leq m, \\ \pi_0(X_k), & k \geq m+1, \end{cases}$$

with $\Pi^{(0)} \equiv \pi_0$ and $\Pi^{(\infty)} \equiv \pi_1$. By construction, both $\Pi^{(m)}$ and $\Pi^{(m-1)}$ use π_1 for stages $1, \dots, m-1$ and π_0 after m , so the only difference is the action chosen at stage m .

For notational convenience, we introduce the pathwise (random) payoff

$$\tilde{J}(\pi) = \delta(\tau_{\pi})g(X_{\tau_{\pi}}) + \lambda \sum_{k=1}^{\tau_{\pi}} \delta(k) H(\pi(X_k))$$

so that J satisfies

$$J^{\pi}(x) = \mathbb{E}_x[\tilde{J}(\pi)].$$

This allows us to perform the event-wise comparison at the level of the random variables $\tilde{J}(\Pi^{(m)})$ and $\tilde{J}(\Pi^{(m-1)})$ before taking expectations to get the result for J . Since we must have $\tau_{\Pi^{(m)}} \geq \tau_{\Pi^{(m-1)}}$ almost surely, we have 4 possible cases.

Case 1: $A = \{\tau_{\Pi^{(m-1)}} \leq m-1\} \cap \{\tau_{\Pi^{(m)}} \leq m-1\}$. In this event both hybrid policies have already stopped strictly before the single time m at which they differ. Thus the realised payoffs are identical, and

$$\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)})\right) 1_A = 0$$

almost surely.

Case 2: $B = \{\tau_{\Pi^{(m-1)}} \geq m+1\} \cap \{\tau_{\Pi^{(m)}} \geq m+1\}$. From time $m+1$ onward, both policies have stopping probability π_0 , so their continuation values are the same. At time m , however, $\Pi^{(m)}$ uses $\pi_1(X_m)$ while $\Pi^{(m-1)}$ uses $\pi_0(X_m)$, contributing only the entropy term. Hence on B ,

$$\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)})\right) 1_B = \delta(m) \lambda (H(\pi_1(X_m)) - H(\pi_0(X_m))) 1_B$$

almost surely.

Case 3: $C = \{\tau_{\Pi^{(m-1)}} = m\} \cap \{\tau_{\Pi^{(m)}} = m\}$. Once again, both policies stop at time m and obtain $\delta(m)g(X_m)$, meaning the only difference at time m is the entropy term. Thus on C ,

$$\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)})\right) 1_C = \delta(m) \lambda (H(\pi_1(X_m)) - H(\pi_0(X_m))) 1_C$$

almost surely.

Case 4: $D = \{\tau_{\Pi^{(m-1)}} = m\} \cap \{\tau_{\Pi^{(m)}} \geq m+1\}$. In this case, $\Pi^{(m-1)}$ stops at m but $\Pi^{(m)}$ continues, which occurs with probability $\pi_0(X_m) - \pi_1(X_m) \geq 0$. The payoff under $\Pi^{(m-1)}$ when this occurs is

$$\tilde{J}(\Pi^{(m-1)}) 1_D = \left(\delta(m)g(X_m) + \lambda \sum_{k=1}^{m-1} \delta(k)H(\pi_1(X_k)) + \lambda \delta(m)H(\pi_0(X_m)) \right) 1_D$$

almost surely.

Under $\Pi^{(m)}$, the process continues and from time $m+1$ onward follows π_0 , yielding

$$\begin{aligned} \tilde{J}(\Pi^{(m)}) 1_D &= \left(\lambda \sum_{k=1}^m \delta(k)H(\pi_1(X_k)) + \mathbb{E}_x \left(\delta(m+\tau)g(X_{m+\tau}) + \lambda \sum_{k=1}^{\tau} \delta(m+k)H(\pi_0(X_{m+k})) \middle| \tau \geq 1 \right) \right) 1_D \\ &\geq \left(\lambda \sum_{k=1}^m \delta(k)H(\pi_1(X_k)) + \delta(m) \mathbb{E}_x \left(\delta(\tau)g(X_{m+\tau}) + \lambda \sum_{k=1}^{\tau} \delta(k)H(\pi_0(X_{m+k})) \middle| \tau \geq 1 \right) \right) 1_D \\ &= \left(\lambda \sum_{k=1}^m \delta(k)H(\pi_1(X_k)) + \delta(m) J^{\pi_0}(X_m) \right) 1_D \end{aligned}$$

almost surely, where the inequality uses the decreasing impatience property. Therefore, on D ,

$$\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)})\right) 1_D \geq \delta(m) \left(J^{\pi_0}(X_m) - g(X_m) + \lambda (H(\pi_1(X_m)) - H(\pi_0(X_m))) \right) 1_D$$

almost surely.

Combining the four cases, noting that $A^c = B \cup C \cup D = \{\tau_{\Pi^{(m-1)}} \geq m\} \cap \{\tau_{\Pi^{(m)}} \geq m\}$ and taking conditional expectations, we have that

$$\begin{aligned}\mathbb{P}(B \mid X_m, A^c) &= 1 - \pi_0(X_m) \\ \mathbb{P}(C \mid X_m, A^c) &= \pi_1(X_m) \\ \mathbb{P}(D \mid X_m, A^c) &= \pi_0(X_m) - \pi_1(X_m).\end{aligned}$$

Finally, from the tower property, we have that for all $m \geq 1$:

$$\begin{aligned}J^{\Pi^{(m)}}(x) - J^{\Pi^{(m-1)}}(x) &= \mathbb{E}_x \left[\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)}) \right) (1_A + 1_B + 1_C + 1_D) \right] \\ &= 0 + \mathbb{E}_x \left[\mathbb{E} \left[\left(\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)}) \right) (1_B + 1_C + 1_D) \mid X_m, A^c \right] \right] \\ &= \mathbb{E}_x \left[\mathbb{P}(B \mid X_m, A^c) \mathbb{E} \left[\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)}) \mid X_m, B \right] \right. \\ &\quad \left. + \mathbb{P}(C \mid X_m, A^c) \mathbb{E} \left[\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)}) \mid X_m, C \right] \right. \\ &\quad \left. + \mathbb{P}(D \mid X_m, A^c) \mathbb{E} \left[\tilde{J}(\Pi^{(m)}) - \tilde{J}(\Pi^{(m-1)}) \mid X_m, D \right] \right] \\ &\geq \mathbb{E}_x \left[\delta(m) \left((\pi_0(X_m) - \pi_1(X_m)) (J^{\pi_0}(X_m) - g(X_m)) + \lambda (H(\pi_1(X_m)) - H(\pi_0(X_m))) \right) \right] \\ &= \mathbb{E}_x \left[\delta(m) \left(U_\lambda(X_m, \pi_1, \pi_0) - U_\lambda(X_m, \pi_0, \pi_0) \right) \right] \\ &\geq 0,\end{aligned}$$

with the last inequality coming from the fact that $\pi_1 = \arg \max_\pi U_\lambda(x, \pi, \pi_0)$. This gives us that $J^{\Pi^{(m)}}(x) \geq J^{\Pi^{(m-1)}}(x)$ almost surely for all m . Applying this inequality successively from $m = 1$ to $m \rightarrow \infty$, the dominated convergence theorem gives us

$$J^{\pi_0}(x) = J^{\Pi^{(0)}}(x) \leq \dots \leq J^{\Pi^{(m-1)}}(x) \leq J^{\Pi^{(m)}}(x) \leq J^{\Pi^{(m+1)}}(x) \leq \dots \leq J^{\Pi^{(\infty)}}(x) = J^{\pi_1}(x)$$

as required. Since x was arbitrary, we have shown $J^{\pi_0}(x) \leq J^{\pi_1}(x)$ for all x .

For the final claim, the best response $\pi^*(x) = \arg \max_\pi U_\lambda(x, \pi, \tilde{\pi})$ is strictly decreasing in the continuation value $J^{\tilde{\pi}}(\cdot)$, so

$$J^{\pi_1}(\cdot) \geq J^{\pi_0}(\cdot) \Rightarrow \pi_2(\cdot) = \arg \max_\pi U_\lambda(\cdot, \pi, \pi_1) \leq \arg \max_\pi U_\lambda(\cdot, \pi, \pi_0) = \pi_1(\cdot). \quad \square$$

Theorem 4.1 (Convergence of the fixed-point iteration). *Let $\pi_1(x) = \arg \max_\pi U_\lambda(x, \pi, \pi_0) \leq \pi_0(x)$ for all $x \in \mathbb{X}$, and define*

$$\pi_\infty(x) = \lim_{n \rightarrow \infty} \pi_n(x)$$

Then π_∞ is an equilibrium.

Proof. By Lemma 4.1, we have $J^{\pi_n}(\cdot) \geq J^{\pi_{n-1}}(\cdot)$ for all $n \geq 1$, and thus (π_n) is pointwise non-increasing and bounded from below by 0, so $\pi_n \downarrow \pi_\infty$ pointwise.

Under the common coupling (same X and (U_k) for all n), the randomised stopping times satisfy $\tau_{\pi_n} \uparrow \tau_{\pi_\infty}$ almost surely. With g bounded and $\delta(k) \rightarrow 0$, dominated convergence implies

$$J^{\pi_n}(x) \uparrow J^{\pi_\infty}(x) \quad \text{for all } x \in \mathbb{X}.$$

Moreover, for each x the one-step maximiser has the continuous closed form

$$\pi_{n+1}(x) = \Phi(J^{\pi_n}(x)), \quad \Phi(z) = \frac{1}{1 + \exp((z - g(x))/\lambda)}.$$

Passing to the limit $n \rightarrow \infty$ yields

$$\pi_\infty(x) = \lim_{n \rightarrow \infty} \pi_{n+1}(x) = \Phi(J^{\pi_\infty}(x)) = \arg \max_{\pi} U_\lambda(x, \pi, \pi_\infty),$$

so π_∞ is a pointwise best response to itself and hence an equilibrium. $\square$

Corollary 4.1 (Existence of an equilibrium). *An equilibrium to the discrete-time time-inconsistent problem exists. In particular, taking $\pi_0 \equiv 1$ yields for every $x \in \mathbb{X}$*

$$\pi_\infty(x) \leq \dots \leq \pi_1(x) \leq \pi_0(x) = 1$$

and $\pi_\infty(x)$ is an equilibrium.

Proof. By construction, we have $\pi_1(x) \leq \pi_0(x) = 1$ for every $x \in \mathbb{X}$. Then we can apply Theorem 4.1 to get the result. $\square$

4.2 Reinforcement Learning Algorithm

Following Theorem 4.1, we can use an iterative method to solve the discrete-time time-inconsistent stopping problem by starting with $\pi_0 \equiv 1$ and alternating between:

- **Policy evaluation:** Given π_n , compute

$$J^{\pi_n}(x) = \mathbb{E}_x \left(\delta(\tau)g(X_\tau) + \lambda \sum_{k=1}^{\tau} \delta(k)H(\pi_n(X_k)) \mid \tau \geq 1 \right).$$

- **Policy update:** Given $J^{\pi_n}(x)$, find

$$\pi_{n+1}(x) = \frac{1}{1 + \exp((J^{\pi_n}(x) - g(x))/\lambda)}$$

From Corollary 4.1, this ensures that we reach an equilibrium $\pi^*(x)$ as $n \rightarrow \infty$. In practice, it can be difficult to compute $J^{\pi_n}(x)$ theoretically. To handle this, we approximate the value function using truncation and regression with parameters ϕ . Since $\delta(k) \downarrow 0$ and g is bounded, this error term is negligible once the horizon cap K is sufficiently large ($\delta(K)$ is sufficiently small). Operationally, the K primarily controls the bias of the solution, while the number of independent rollouts per starting state R controls the variance, and so these parameters can be tuned separately when fitting J_ϕ .

Once again, we will be using a fully connected neural network to represent the value function, however any parameterised system can be used. We will also use a Monte Carlo method to sample the data. Following the above setup, Algorithm 3 presents a numerical implementation to solve the discrete-time time-inconsistent stopping problem.

Algorithm 3 Entropy-regularised RL for discrete-time time-inconsistent stopping

-
- 1: **Inputs:** Discount δ , payoff g , exploration weight $\lambda > 0$, learning rate η , iterations N , batch size M , rollouts per state R , horizon cap K
- 2: **Initialise:** Policy $\pi_0(x) \equiv 1$, network parameters ϕ
- 3: **for** $n = 0$ to $N - 1$ **do**
- 4: **State sampling:** Draw starting states $\{x^{(m)}\}_{m=1}^M$.
- Monte Carlo for** $J(\cdot; \pi_n)$:
- 5: **for** $m = 1$ to M **do**
- 6: **for** $r = 1$ to R **do**
- 7: Sample one step $y^{(m,r)} \sim Q(x^{(m)}, \cdot)$
- 8: Simulate a trajectory $\{X_k^{(m,r)}\}_{k \geq 1}$ with $X_1^{(m,r)} = y^{(m,r)}$ under *fixed* policy π_n
- 9: At step $k \geq 1$, stop with probability $\pi_n(X_k^{(m,r)})$
- 10: Let $\tau^{(m,r)} \in \{1, \dots, K\}$ be the corresponding stopping time.
- 11: Compute the MC target

$$\widehat{J}^{(m,r)} = \delta(\tau^{(m,r)}) g(X_{\tau^{(m,r)}}^{(m,r)}) + \lambda \sum_{k=1}^{\tau^{(m,r)}} \delta(k) H(\pi_n(X_k^{(m,r)})).$$

- 12: Aggregate target $\bar{J}^{(m)} \leftarrow \frac{1}{R} \sum_{r=1}^R \widehat{J}^{(m,r)}$
- 13: **TD residual for continuation:** $\varepsilon^{(m)} \leftarrow \bar{J}^{(m)} - J_\phi(x^{(m)})$
- 14: **Regression step (update J_ϕ):**

$$\phi \leftarrow \phi - \eta \nabla_\phi \sum_{m=1}^M \left(\varepsilon^{(m)} \right)^2$$

- 15: **Policy update (best response to π_n):** for each x in the training batch, set

$$\pi_{n+1}(x) \leftarrow \frac{1}{1 + \exp([J_\phi(x) - g(x)]/\lambda)}.$$

- 16: **Value reporting:**

$$V_n(x) = U_\lambda(x, \pi_n(x), \pi_n) = \pi_n(x) g(x) + (1 - \pi_n(x)) J_\phi(x) + \lambda H(\pi_n(x))$$

- 17: **Convergence check:** Stop if $\max_{x \in \text{batch}} |\pi_{n+1}(x) - \pi_n(x)| \leq \text{tol}$.
- 18: **Return:** equilibrium policy estimate $\pi_\infty \approx \pi_N$ and trained J_ϕ .
-

5 Time-Inconsistent Stopping – Continuous-Time Case

We consider a time-inconsistent optimal stopping problem for a one-dimensional diffusion state process. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space supporting a standard Brownian motion W , and fix $T \in (0, \infty]$. The state $X = (X_t)_{t \geq 0}$ evolves as the time-homogeneous diffusion

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad X_0 = x \quad (36)$$

where $b, \sigma : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the usual Lipschitz and linear-growth conditions to ensure a unique strong solution exists. We write $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ for the natural filtration and denote

$$\mathcal{L} = \frac{1}{2} \sigma^2(x) \partial_{xx} + b(x) \partial_x$$

as the infinitesimal generator of X .

The payoff function $g \geq 0$ is still bounded, however in this setup we require it to be continuous on $\mathbb{R}$. Moreover, the strictly decreasing discount function $\delta : [0, \infty) \rightarrow [0, 1]$ now satisfies:

1. $\delta \in C^1([0, \infty))$.
2. $\delta(0) = 1$.
3. $\lim_{t \rightarrow \infty} \delta(t) = 0$.
4. The right derivative $\delta'(0^+)$ exists.
5. $\delta(s)\delta(t) \leq \delta(s+t)$ for all $s, t \geq 0$.

Given (t, x) with $0 \leq t < T$, the objective is to find the optimal $\tau \in \mathcal{T}_{[t, T]}$ that maximises the discounted payoff

$$\widehat{V}^\tau(t, x) = \mathbb{E}_{t, x} [\delta(\tau - t) g(X_\tau)].$$

Similarly to the discrete-time case, we can represent the optimal policy as a Borel set $S \subseteq [0, T] \times \mathbb{R}$, where $(t, x) \in S$ indicates the agent should stop at time t in state x .

Given that continuous-time problems allow for instantaneous deviations, our definition of an equilibrium must exclude any profitable short-horizon spikes. For this section, we will focus on finding a weak equilibrium to the time-inconsistent stopping problem. We will follow the definition provided in [13] and [34]. Given that S is admissible and $(t, x) \in [0, T] \times \mathbb{R}$ (where $X_t = x$), let

$$\rho(t, x, S) = \inf \{ s \geq t : (s, X_s) \in S \} \wedge T$$

be the first hitting time of S after t , and for $\varepsilon > 0$, define

$$\rho^\varepsilon(t, x, S) = \inf \{ s \geq t + \varepsilon : (s, X_s) \in S \} \wedge T,$$

as the post-deviation hitting time after continuing over $(t, t + \varepsilon)$. Similarly to section 4, we work with equilibria to solve the problem.

Definition 5.1. We say that S is a *weak equilibrium* if

$$g(x) \leq \widehat{V}^{\rho(t,x,S)}(t,x) \quad (37)$$

for all $(t,x) \notin S$, and

$$\liminf_{\varepsilon \downarrow 0} \frac{g(x) - \widehat{V}^{\rho^\varepsilon(t,x,S)}(t,x)}{\varepsilon} \geq 0 \quad (38)$$

for all $(t,x) \in S$. See also [13] and [34] for definitions on mild, strong and almost strong equilibria.

When $T = \infty$, it makes sense to work with time-independent stopping regions $S \subseteq \mathbb{R}$ instead, which produces the corresponding stopping time

$$\tau_S = \inf\{t \geq 0 : X_t \in S\}.$$

The corresponding value is thus $\widehat{V}^{\tau_S}(t,x) = \mathbb{E}_{t,x}[\delta(\tau_S - t)g(X_{\tau_S})]$.

5.1 Application to Reinforcement Learning

In our reinforcement learning framework, smoothing the stop/continue choice is once again crucial for learning and equilibrium analysis. Instead of a stopping probability on each discrete step, we introduce a *stopping intensity* (rate) $\alpha = (\alpha(X_s))_{s \geq 0}$, taking values in a bounded interval $[0, K]$ with $K < \infty$ (here, we will be considering the infinite time horizon case, $T = \infty$). In a similar style to [7], this now transforms the problem into a stochastic control problem, for which we will now develop the theory.

Given a measurable $\alpha : \mathbb{R} \rightarrow [0, K]$, define the inhomogeneous Poisson (Cox) clock with compensator $\Lambda_s = \int_0^s \alpha(X_r) dr$, and set the induced stopping time as the first-jump time

$$\tau_\alpha = \inf\{s \in [0, T] : \Delta N_s^\alpha = 1\} = \inf\{s \in [0, T] : \Lambda_s \geq E\},$$

where $E \sim \text{Exp}(1)$ is independent of $(X, \mathbb{F})$. This construction admits a convenient interpretation: given the realised state path $(X_s)_{s \geq 0}$, the induced stopping time τ_α has the conditional survival function

$$\mathbb{P}(\tau_\alpha \geq s \mid \mathcal{F}_s) = \exp\left(-\int_0^s \alpha(X_r) dr\right),$$

and hence conditional density

$$\mathbb{P}(\tau_\alpha \in [s, s+ds] \mid \mathcal{F}_s) = \alpha(X_s) \exp\left(-\int_0^s \alpha(X_r) dr\right) ds.$$

Equivalently, given survival up to time s , the probability of stopping in the next infinitesimal interval is

$$\mathbb{P}(\tau_\alpha \in [s, s+ds] \mid \tau_\alpha \geq s, \mathcal{F}_s) = \alpha(X_s) ds.$$

This is the continuous-time analogue of replacing a hard stop/continue decision by a smooth randomised rule, with the bound K ensuring the stopping mechanism remains well-behaved.

From this setup, the expected discounted payoff at $t = 0$ then reads

$$\begin{aligned} V^\alpha(0, x) &= \mathbb{E}_x[\delta(\tau_\alpha)g(X_{\tau_\alpha})] \\ &= \mathbb{E}_x\left[\int_0^\infty \delta(s)g(X_s)\alpha(X_s)\exp\left(-\int_0^s \alpha(X_r)dr\right)ds\right] \\ &= \mathbb{E}_x\left[\int_0^\infty \delta(s)g(X_s)\alpha(X_s)p_s ds\right], \end{aligned} \quad (39)$$

where in the last line we have once again used the survival process $p_s = \mathbb{P}(\tau_\alpha \geq s) = \exp\left(-\int_0^s \alpha(X_r)dr\right)$ that satisfies

$$dp_s = -\alpha(X_s)p_s ds \quad (40)$$

with $p_0 = 1$, similarly to [7]. This motivates introducing p as a free parameter to get:

$$\tilde{V}^\alpha(t, x, p) = \mathbb{E}_{t,x,p}\left[\int_t^\infty \delta(s)g(X_s)\alpha(X_s)p_s ds\right]$$

for any (t, x, p) . Note that if $t = 0$ and $p_0 = 1$, then the solution of (40) is $p_s = \exp\left(-\int_0^s \alpha(X_r)dr\right)$, meaning that $\tilde{V}^\alpha(0, x, 1) = V^\alpha(0, x)$ by (39). Extending on this relationship, we will further define

$$\begin{aligned} V^\alpha(t, x) &= \mathbb{E}_{t,x}[\delta(\tau_\alpha)g(X_{\tau_\alpha}) \mid \tau_\alpha \geq t] \\ &= \tilde{V}^\alpha(t, x, 1) \end{aligned}$$

for any (t, x) . This also gives us the property that

$$pV^\alpha(t, x) = \tilde{V}^\alpha(t, x, p) \quad (41)$$

for any (t, x, p) . The augmented value $\tilde{V}^\alpha$ is Markovian in (X_t, p_t) , making it amenable to the DPP. This formulation will become useful when deriving the HJB equation for the problem.

Definition 5.2. We say that $\alpha : \mathbb{R} \rightarrow [0, K]$ is *admissible* (denoted $\alpha \in \mathcal{A}$) if it is measurable and $V^\alpha(0, x)$ in (39) is finite.

This mirrors the regularity conditions imposed on stopping regions in the non-reinforcement learning formulation. We now redefine the concept of a weak equilibrium:

Definition 5.3. Let α be a candidate rate and fix $(t, x) \in [0, \infty) \times \mathbb{R}$. For $\varepsilon > 0$ and $a \in [0, K]$, define the *spike deviation* on the short interval $[t, t + \varepsilon]$ by

$$(\alpha^{t,\varepsilon,a})_s = \begin{cases} a, & s \in [t, t + \varepsilon) \\ \alpha(X_s), & s \geq t + \varepsilon. \end{cases}$$

At $t = 0$, we say that α^* is a *weak equilibrium rate* if

$$\liminf_{\varepsilon \downarrow 0} \frac{V^{\alpha^*}(0, x) - V^{\alpha^{0,\varepsilon,a}}(0, x)}{\varepsilon} \geq 0 \quad (42)$$

for all $a \in [0, K]$ and all x . We will expand on this further in the following section, however this corresponds to α^* being the first-order best response to itself.

5.2 Weak Equilibrium Rate HJB Equation

Fix an admissible candidate rate α and a point (t, x) . We now derive and analyse the weak equilibrium equation for the continuous-time time-inconsistent stopping problem with a reinforcement learning framework using the DPP. We follow a similar method as [33] and Theorem 2.2, and once again use the equilibrium definition outlined in [13].

To start, we will work with $\tilde{V}^\alpha(t, x, p)$ over the short time horizon $[t, t + \varepsilon]$. A first-order decomposition of $\tilde{V}^\alpha(t, x, p)$ gives us

$$\begin{aligned}\tilde{V}^\alpha(t, x, p) &= \mathbb{E}_{t,x,p} \left[\int_t^{t+\varepsilon} \delta(s) g(X_s) \alpha(X_s) p_s ds + \int_{t+\varepsilon}^\infty \delta(s) g(X_s) \alpha(X_s) p_s ds \right] \\ &= \varepsilon \delta(t) g(x) \alpha(x) p + \mathbb{E}_{t,x,p} \left[\tilde{V}^\alpha(t + \varepsilon, X_{t+\varepsilon}, p_{t+\varepsilon}) \right] + o(\varepsilon).\end{aligned}$$

By (41), we have $\tilde{V}^\alpha(t + \varepsilon, X_{t+\varepsilon}, p_{t+\varepsilon}) = p_{t+\varepsilon} V^\alpha(t + \varepsilon, X_{t+\varepsilon})$. Assuming V^α is sufficiently smooth⁵, we apply Itô's formula to the product $p_s V^\alpha(s, X_s)$ on $[t, t + \varepsilon]$ (note that once again p_s has finite variation):

$$\begin{aligned}d(p_s V^\alpha(s, X_s)) &= p_s dV^\alpha(s, X_s) + V^\alpha(s, X_s) dp_s \\ &= p_s ((\partial_t + \mathcal{L})V^\alpha(s, X_s)) ds + p_s \partial_x V^\alpha(s, X_s) \sigma(X_s) dW_s - \alpha(X_s) p_s V^\alpha(s, X_s) ds.\end{aligned}$$

Taking expectations and using a first-order expansion yields

$$\mathbb{E}_{t,x,p}[p_{t+\varepsilon} V^\alpha(t + \varepsilon, X_{t+\varepsilon})] = p V^\alpha(t, x) + \varepsilon p \left((\partial_t + \mathcal{L})V^\alpha(t, x) - \alpha(x) V^\alpha(t, x) \right) + o(\varepsilon).$$

Substituting back into the DPP expansion and using $\tilde{V}^\alpha(t, x, p) = p V^\alpha(t, x)$ gives

$$p V^\alpha(t, x) = \varepsilon \delta(t) g(x) \alpha(x) p + p V^\alpha(t, x) + \varepsilon p \left((\partial_t + \mathcal{L})V^\alpha(t, x) - \alpha(x) V^\alpha(t, x) \right) + o(\varepsilon).$$

Cancelling $p V^\alpha(t, x)$, dividing by $\varepsilon p > 0$, and letting $\varepsilon \rightarrow 0$ yields

$$(\partial_t + \mathcal{L})V^\alpha(t, x) + \alpha(x) (\delta(t) g(x) - V^\alpha(t, x)) = 0. \quad (43)$$

In particular, at $t = 0$:

$$(\partial_t + \mathcal{L})V^\alpha(0, x) + \alpha(x) (g(x) - V^\alpha(0, x)) = 0. \quad (44)$$

We will now prove the existence and uniqueness of solutions to the equation with the following lemma. Note that if α is not continuous in x , then V^α is not necessarily $C^{1,2}([0, \infty) \times \mathbb{R})$. We will thus look for solutions in the Sobolev space

$$W_p^{1,2}([0, \infty) \times \mathbb{R}) = \{u \in L^p([0, \infty) \times \mathbb{R}) : \partial_t u, \partial_x u, \partial_{xx} u \in L^p([0, \infty) \times \mathbb{R})\}$$

with $1 < p < \infty$.

⁵This can be justified by first treating the case of continuous α , which gives $V^\alpha \in C^{1,2}$ by the remark after Lemma 5.1, then extending to general admissible α by approximating α with smooth α^n and passing to the limit in $W_p^{1,2}$. See also [17].

Lemma 5.1. *Let α be an admissible rate. Then the linear HJB equation (43) has a unique solution $V^\alpha(\cdot, \cdot) \in W_p^{1,2}([0, \infty) \times \mathbb{R})$.*

Proof. This proof is similar to the proof of Theorem 3.1 in [7], except that we now work with weak Sobolev solutions.

Firstly, note that $\mathcal{L}$ is uniformly parabolic and that δ , α , and g are all bounded. By the theory for linear parabolic equations in L^p on the whole half-space derived in chapters 2, 4 and 5 in [17], there exists a weak solution

$$V^\alpha \in W_p^{1,2}([0, \infty) \times \mathbb{R})$$

to (43). Moreover, this solution is uniquely determined among all $W_p^{1,2}$ solutions satisfying $V^\alpha(t, x) \rightarrow 0$ as $t \rightarrow \infty$.

For uniqueness, suppose V and V' are two such solutions. Then $w = V - V'$ satisfies

$$(\partial_t + \mathcal{L})w(t, x) - \alpha(x)w(t, x) = 0.$$

Since $\alpha(x)$ is bounded and the coefficients of $\mathcal{L}$ are continuous with (at most) linear growth, this equation is uniformly parabolic. Finally, since $-\alpha(x) \leq 0$, the parabolic maximum principle derived in chapter 2 of [18] implies that any interior extremum of w propagates forward in time. Combined with $w(t, x) \rightarrow 0$ as $t \rightarrow \infty$, we conclude that $w \equiv 0$, that is, $V = V'$. $\square$

Remark. If α is continuous in x , then a similar argument as above can be used to show that $V^\alpha(\cdot, \cdot) \in C^{1,2}([0, \infty) \times \mathbb{R})$, since all of the coefficients are now continuous and bounded.

We will now analyse the weak equilibrium condition.

A direct expansion of the numerator in (42) and a similar analysis as above yields

$$\begin{aligned} V^\alpha(0, x) - V^{\alpha^{0, \varepsilon, a}}(0, x) &= \mathbb{E}_x \left[\int_0^\varepsilon \delta(s) g(X_s) \left(\alpha(X_s) \exp \left(- \int_0^s \alpha(X_r) dr \right) - a \exp(-as) \right) ds \right] \\ &\quad + \mathbb{E}_x \left[\left(\tilde{V}^\alpha(\varepsilon, X_\varepsilon, p_\varepsilon^\alpha) - \tilde{V}^\alpha(\varepsilon, X_\varepsilon, p_\varepsilon^a) \right) \right] + o(\varepsilon) \\ &= \varepsilon(\alpha(x) - a)g(x) + \mathbb{E}_x [V^\alpha(\varepsilon, X_\varepsilon)(p_\varepsilon^\alpha - p_\varepsilon^a)] + o(\varepsilon) \\ &= \varepsilon(\alpha(x) - a)g(x) + V^\alpha(0, x)((1 - \varepsilon\alpha(x)) - (1 - a\varepsilon)) + o(\varepsilon) \\ &= \varepsilon(\alpha(x) - a)g(x) - \varepsilon(\alpha(x) - a)V^\alpha(0, x) + o(\varepsilon) \\ &= \varepsilon(\alpha(x) - a)(g(x) - V^\alpha(0, x)) + o(\varepsilon). \end{aligned}$$

Thus, after dividing by ε and taking $\liminf_{\varepsilon \rightarrow 0}$, the weak equilibrium condition (42) is equivalent to

$$\alpha^*(x) \in \arg \max_{\alpha \in [0, K]} \alpha(g(x) - V^{\alpha^*}(0, x)) = \begin{cases} K, & g(x) > V^{\alpha^*}(0, x), \\ 0, & g(x) < V^{\alpha^*}(0, x), \\ \text{any } a \in [0, K], & g(x) = V^{\alpha^*}(0, x). \end{cases} \quad (45)$$

In other words, the agent increases the stopping intensity to the maximum rate when the immediate payoff exceeds the continuation value, decreases it to 0 when the continuation value is higher, and

is indifferent on the set $g(x) = V^{\alpha^*}(0, x)$. This equation, when combined with (43), can be solved to find $(V^*, \alpha^*) = (V^{\alpha^*}, \alpha^*)$ that maximises (39) for all (t, x) . We will make this rigorous with the following theorem.

Theorem 5.1 (Verification Theorem). *Suppose that $(V^*, \alpha^*) \in W_p^{1,2}([0, \infty) \times \mathbb{R}) \times \mathcal{A}$ satisfies*

$$(\partial_t + \mathcal{L}) V^*(t, x) + \alpha^*(x) (\delta(t)g(x) - V^*(t, x)) = 0 \quad (46)$$

for any (t, x) , and

$$\alpha^*(x) = \begin{cases} K, & g(x) > V^*(0, x), \\ 0, & g(x) < V^*(0, x), \\ \text{any } a \in [0, K], & g(x) = V^*(0, x) \end{cases} \quad (47)$$

for any x . Then α^* is a weak equilibrium rate and $V^*(0, x) = V^{\alpha^*}(0, x)$.

Proof. We split the argument into two steps: identification of the value function and verification of the weak equilibrium condition.

Step 1: $V^* = V^{\alpha^*}$. From our derivation above, we have that for any fixed α , the corresponding value V^α satisfies (44) in the weak sense. In our case, after setting $\alpha = \alpha^*$ we have

$$(\partial_t + \mathcal{L})V^{\alpha^*}(t, x) + \alpha^*(x) (\delta(t)g(x) - V^{\alpha^*}(t, x)) = 0,$$

that is, both V^* and V^{α^*} solve the same HJB equation. Finally, by the uniqueness of the weak Sobolev solution from Lemma 5.1, this must mean that $V^* = V^{\alpha^*}$, proving the identification.

Step 2: α^* is a weak equilibrium rate. Fix $x \in \mathbb{R}$ and $a \in [0, K]$, and consider the spike variation $\alpha^{*0, \varepsilon, a}$. Since $V^* = V^{\alpha^*}$, the first-order expansion derived earlier gives

$$V^{\alpha^*}(0, x) - V^{\alpha^{*0, \varepsilon, a}}(0, x) = \varepsilon (\alpha^*(x) - a) (g(x) - V^{\alpha^*}(0, x)) + o(\varepsilon),$$

for almost every x . Dividing by ε and taking $\liminf_{\varepsilon \downarrow 0}$ yields

$$\liminf_{\varepsilon \downarrow 0} \frac{V^{\alpha^*}(0, x) - V^{\alpha^{*0, \varepsilon, a}}(0, x)}{\varepsilon} = (\alpha^*(x) - a)(g(x) - V^{\alpha^*}(0, x)).$$

By the selection rule (47), we must have

$$(\alpha^*(x) - a)(g(x) - V^{\alpha^*}(0, x)) \geq 0 \quad \forall a \in [0, K],$$

for almost every x , and therefore the weak equilibrium condition (42) holds. $\square$

5.3 Reinforcement Learning Algorithm

From Theorem 5.1, we now have an algorithm to iteratively find a weak equilibrium rate and its corresponding value (V^*, α^*) , starting with $\alpha^{(0)} \equiv K$:

- **Policy evaluation:** Given the current policy $\alpha^{(n)}(x)$, find $V^{(n)}(t, x)$ by solving

$$(\partial_t + \mathcal{L}) V^{(n)}(t, x) + \alpha^{(n)}(x) (\delta(t)g(x) - V^{(n)}(t, x)) = 0. \quad (48)$$

- **Policy update:** Update the intensity by the weak equilibrium selection rule

$$\alpha^{(n+1)}(x) = \begin{cases} K, & g(x) \geq V^{(n)}(0, x), \\ 0, & g(x) < V^{(n)}(0, x). \end{cases} \quad (49)$$

We also note here that in the case that $g(x) = V^{(n)}(0, x)$, we set $\alpha(x) = K$.

We will now prove the convergence of this algorithm. To do so, we will first need to make use of the decreasing impatience property to work with δ .

Lemma 5.2 (Value comparison under DI). *Assume $\delta(s+t) \geq \delta(s)\delta(t)$ for all $s, t \geq 0$. Then for any admissible rate α and all $(t, x) \in [0, \infty) \times \mathbb{R}$,*

$$V^\alpha(t, x) \geq \delta(t) V^\alpha(0, x). \quad (50)$$

In particular, if $\alpha^{(n)} \leq \alpha^{(n-1)}$ and $(\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(g(x) - V^{(n-1)}(0, x)) \leq 0$, we have that

$$(\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(\delta(t)g(x) - V^{(n-1)}(t, x)) \leq 0 \quad (51)$$

Proof. Using the survival representation (39), for $t \geq 0$,

$$V^\alpha(t, x) = \mathbb{E}_{t,x} \left[\int_t^\infty \delta(s)g(X_s)\alpha(X_s) \exp\left(-\int_t^s \alpha(X_r)dr\right) ds \right].$$

Now, from the decreasing impatience property of δ and the fact that X is a Markov process, we have that

$$\begin{aligned} V^\alpha(t, x) &\geq \delta(t) \mathbb{E}_{t,x} \left[\int_t^\infty \delta(s-t)g(X_s)\alpha(X_s) \exp\left(-\int_t^s \alpha(X_r)dr\right) ds \right] \\ &= \delta(t) \mathbb{E}_{t,x} \left[\int_0^\infty \delta(u)g(X_{t+u})\alpha(X_{t+u}) \exp\left(-\int_t^{t+u} \alpha(X_r)dr\right) du \right] \\ &= \delta(t) \mathbb{E}_x \left[\int_0^\infty \delta(u)g(X_u)\alpha(X_u) \exp\left(-\int_0^u \alpha(X_r)dr\right) du \right] \\ &= \delta(t) V^\alpha(0, x) \end{aligned}$$

where we have made the substitution $u = s - t$. This proves the first claim.

For the second claim, note that since the first term $\alpha^{(n-1)}(x) - \alpha^{(n)}(x) \geq 0$, we can use (50) to get $(\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(\delta(t)g(x) - V^{(n-1)}(t, x)) \leq (\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(\delta(t)g(x) - \delta(t)V^{(n-1)}(0, x))$.

After dividing by $\delta(t) \leq 1$, we have

$$(\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(\delta(t)g(x) - V^{(n-1)}(t, x)) \leq (\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(g(x) - V^{(n-1)}(0, x)) \leq 0$$

As required. $\square$

Theorem 5.2 (Convergence). *Suppose that $\alpha^{(0)} \equiv K$ for all (t, x) . Then, following the iterative procedure outlined in (48) and (49), we have*

$$V^{(n)} \uparrow V^*, \quad \alpha^{(n)} \downarrow \alpha^*,$$

for all (t, x) as $n \rightarrow \infty$. Moreover, the limit pair (V^*, α^*) solves (43) and (45).

Proof. We write $V^{(n)} = V^{\alpha^{(n)}}$ for the unique weak solution in $W_p^{1,2}([0, \infty) \times \mathbb{R})$ to the full policy evaluation PDE

$$(\partial_t + \mathcal{L})V^{(n)}(t, x) + \alpha^{(n)}(x)(\delta(t)g(x) - V^{(n)}(t, x)) = 0, \quad t > 0, x \in \mathbb{R}, \quad (52)$$

with $V^{(n)}(t, x) \rightarrow 0$ as $t \rightarrow \infty$, whose existence and uniqueness follow from Lemma 5.1. We split the proof into two steps. Step 1 establishes monotonicity, while Step 2 verifies the existence of the limit and equilibrium conditions.

Step 1: $V^{(n+1)} \geq V^{(n)}$ and $\alpha^{(n+1)} \leq \alpha^{(n)}$ for all n . We can prove this using induction. For the base case $n = 0$, since $\alpha^{(0)}$ is defined to be the maximum stopping rate K , we clearly have that $\alpha^{(1)} \leq \alpha^{(0)}$.

Now, for the inductive step, we work on the assumption that $\alpha^{(n)} \leq \alpha^{(n-1)}$. If we subtract (52) at indices n and $n - 1$, with $\Delta V = V^{(n)} - V^{(n-1)}$, we get

$$(\partial_t + \mathcal{L})\Delta V(t, x) + \alpha^{(n)}(x)(\delta(t)g(x) - V^{(n)}(t, x)) - \alpha^{(n-1)}(x)(\delta(t)g(x) - V^{(n-1)}(t, x)) = 0.$$

Adding and subtracting $\alpha^{(n)}(x)V^{(n-1)}(t, x)$ and rearranging gives

$$(\partial_t + \mathcal{L})\Delta V(t, x) - \alpha^{(n)}(x)\Delta V(t, x) = (\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(\delta(t)g(x) - V^{(n-1)}(t, x)). \quad (53)$$

By the update rule (49), we have $\alpha^{(n)}(x) \in \arg \max_{a \in [0, K]} a(g(x) - V^{(n-1)}(0, x))$ (with tie sent to K), meaning

$$(\alpha^{(n-1)}(x) - \alpha^{(n)}(x))(g(x) - V^{(n-1)}(0, x)) \leq 0 \quad \forall x.$$

Thus, by Lemma 5.2, the right-hand side of (53) is also ≤ 0 on $[0, \infty) \times \mathbb{R}$.

Since the zeroth-order coefficient in the left-hand side is $-\alpha^{(n)}(x) \leq 0$, the (weak) parabolic maximum principle in $W_p^{1,2}$ (with the boundary condition $\Delta V(t, \cdot) \rightarrow 0$ as $t \rightarrow \infty$ ensured by Lemma 5.1) implies $\Delta V \geq 0$, that is, $V^{(n)} \geq V^{(n-1)}$ pointwise. For the rates, since $V^{(n)}(0, \cdot) \geq V^{(n-1)}(0, \cdot)$, we have that $\{x : g(x) \geq V^{(n)}(0, x)\} \subseteq \{x : g(x) \geq V^{(n-1)}(0, x)\}$, meaning the bang-bang update yields $\alpha^{(n+1)} \leq \alpha^{(n)}$ pointwise.

Combining these, we have established that $V^{(n)}$ and $\alpha^{(n)}$ are increasing and decreasing respectively in n .

Step 2: Limit to V^* and α^* . The sequences are monotone and bounded: $0 \leq \alpha^{(n)} \leq K$, $0 \leq V^{(n)} \leq V^{(n+1)}$ for all n . By Lemma 5.1, we have that $V^{(n)}(t, x) \rightarrow 0$ as $t \rightarrow \infty$. Similarly to Theorem 2.4 Step 3, we will now show that $\{V^{(n)}\}_{n \geq 0}$ is uniformly bounded on compact cylinders in the local $W_p^{1,2}$ norm, which yields the compactness needed to pass to the limit in (52). Fix $T, R > 0$ and set $Q_{T,R} = (0, T) \times (-R, R)$. Rewriting (52) as

$$(\partial_t + \mathcal{L} - \alpha^{(n)}(x))V^{(n)}(t, x) = -\alpha^{(n)}(x)\delta(t)g(x),$$

the coefficients of $\partial_t + \mathcal{L} - \alpha^{(n)}$ are uniformly bounded on $Q_{T,R}$ (since $0 \leq \alpha^{(n)} \leq K$) and the right-hand side is uniformly bounded in $L^p(Q_{T,R})$ (since δ and g are bounded). Hence, by the standard local L^p a priori estimates for linear parabolic equations in $W_p^{1,2}$, there exists a constant $C_{T,R}$ independent of n such that

$$\sup_{n \geq 0} \|V^{(n)}\|_{W_p^{1,2}(Q_{T,R})} \leq C_{T,R}.$$

Consequently, by the Banach-Alaoglu theorem we can extract a sub-sequence such that $V^{(n)} \rightarrow V^*$ in $W_p^{1,2}(Q_{T,R})$. Following [30], we further obtain that $V^{(n)} \rightarrow V^*$ in $L^p(Q_{T',R'})$ for any $0 < T' < T$, $0 < R' < R$. Thus, there exist pointwise limits

$$V^*(t, x) = \lim_{n \rightarrow \infty} V^{(n)}(t, x) \in [0, \infty), \quad \alpha^*(x) = \lim_{n \rightarrow \infty} \alpha^{(n)}(x) \in [0, K].$$

From the standard stability properties of linear parabolic equations in $W_p^{1,2}$, we can pass the limit in (52) on compact time sets to conclude that⁶

$$(\partial_t + \mathcal{L})V^*(t, x) + \alpha^*(x)(\delta(t)g(x) - V^*(t, x)) = 0. \quad (54)$$

Finally, after letting $n \rightarrow \infty$ in the update rule (49) yields

$$\alpha^*(x) \in \arg \max_{a \in [0, K]} a(g(x) - V^*(0, x)), \quad (55)$$

which are precisely (44) and (45). This completes the proof. $\square$

The convergence result in Theorem 5.2 provides a rigorous justification for viewing the above scheme as a monotone policy iteration algorithm in the space of admissible stopping intensities. From a reinforcement learning perspective, the key here is that fixing the rate α reduces the time-inconsistent problem to a linear policy evaluation equation for V , while the weak equilibrium condition induces a closed form improvement step that depends only on the sign of $g(x) - V^\alpha(0, x)$. The resulting equilibrium rate then interpolates between immediate stopping and full continuation, providing a smooth approximation to the original equilibrium stopping regions established in the beginning of this section.

Similarly to the time-consistent setting, this policy iteration algorithm is well suited to numerical implementation, as each iteration consists of a regression-based policy evaluation step followed by a deterministic policy update. In Algorithm 4, we use a fully connected neural network to represent V . To train the model we sample the underlying dynamics of X in batches of size M and perform gradient descent to tune the parameters θ .

⁶See also [30] for more detailed arguments regarding extracting a subsequence of $V^{(n)}$ and its derivatives, as well as passing the limit in (52).

Algorithm 4 RL policy iteration for continuous-time time-inconsistent stopping

- 1: **Input:** Discount $\delta(\cdot)$, payoff $g(\cdot)$, horizon T , grid size L , batch size M , learning rate η , iterations N , maximum rate K , tie tolerance $\varepsilon_{\text{stop}} > 0$.
- 2: **Initialise:** Time grid $t_l = l\Delta t$, with $\Delta t = T/L$ ($K\Delta t \leq 1$) and $l = 0, \dots, L$. For each slice $l = 0, \dots, L-1$, initialise parameters θ_l for model V_{θ_l} .
- 3: Set terminal value $V_T(x) \leftarrow \delta(T)g(x)$.
- 4: Set previous value $\widehat{V}_0 \leftarrow \text{None}$.
- 5: **for** $n = 1$ to N **do**
- 6: Generate M simulated paths $\{X_{t_l}^{(m)}\}_{l=0}^L$ of the environment and payoffs $g_l^{(m)} \leftarrow g(X_{t_l}^{(m)})$.
- 7: Set, for each m , the one-step continuation target at maturity:

$$V_{\text{next}}^{(m)} \leftarrow V_T(X_{t_L}^{(m)}) = \delta(T)g_L^{(m)}.$$

- 8: **for** $l = L-1$ down to 0 **do**
- 9: **Optimal intensity:** for each path m , set

$$\alpha_{t_l}^{(m)} \leftarrow \begin{cases} K, & \widehat{V}_0 = \text{None}, \\ K, & g_l^{(m)} \geq \widehat{V}_0(X_{t_l}^{(m)}, 0) - \varepsilon_{\text{stop}}, \\ 0, & \text{otherwise.} \end{cases}$$

- 10: **TD residual:** for each m , define

$$\delta_l^{(m)} = \delta(t_l)g_l^{(m)}\alpha_{t_l}^{(m)}\Delta t + (1 - \alpha_{t_l}^{(m)}\Delta t)V_{\text{next}}^{(m)} - V_{\theta_l}(X_{t_l}^{(m)}, t_l).$$

- 11: **Regression step:** minimise $\sum_{m=1}^M (\delta_l^{(m)})^2$ by one gradient step

$$\theta_l \leftarrow \theta_l - \eta \nabla_{\theta_l} \sum_{m=1}^M (\delta_l^{(m)})^2.$$

- 12: **Backward target update:** set the continuation target for the next step

$$V_{\text{next}}^{(m)} \leftarrow V_{\theta_l}(X_{t_l}^{(m)}, t_l), \quad m = 1, \dots, M.$$

- 13: **Policy update:** set $\widehat{V}_0 \leftarrow V_{\theta_0}(\cdot, 0)$.

- 14: **Return:** Trained slice values $\{V_{\theta_l}\}_{l=0}^{L-1}$ and optimal intensity

$$\alpha(x) = K \mathbf{1}\left\{g(x) \geq \widehat{V}_0(x, 0) - \varepsilon_{\text{stop}}, g(x) > 0\right\}.$$

5.4 Numerical Example – American Put

We will now verify the above theory and Algorithm 4 with a numerical example. We revisit the classical American put

$$g(X) = (K_{\text{strike}} - X)^+, \quad dX_t = X_t(r dt + \sigma dW_t)$$

with the same parameters as [3]

$$X_0 = K_{\text{strike}} = 40, \quad r = 0.06, \quad \sigma = 0.40, \quad T = 1,$$

however we will now use the non-exponential discounting factor $\delta(t) = \frac{1}{1+rt}$. We note that this choice satisfies the decreasing impatience property, since

$$\begin{aligned} \delta(s)\delta(t) &= \frac{1}{(1+rs)(1+rt)} \\ &= \frac{1}{1+r(s+t)+r^2st} \\ &\leq \frac{1}{1+r(s+t)} \\ &= \delta(s+t) \end{aligned}$$

for all $s, t \geq 0$. Similarly to subsection 2.4, we discretise the time horizon into $L = 50$ points and use a separate neural network $V_\theta(\cdot, t_l)$ for each slice t_l to learn the model. We further parameterise the value as

$$V_\theta(X_{t_l}, t_l) = g(X_{t_l}) + f_\theta(X_{t_l}, g(X_{t_l}), t_l/T),$$

with f_θ represented as a fully connected network with width 21 and 2 hidden layers. Training is then performed in batch sizes of $M = 2^{10}$ for $N = 3000$ iterations using the Adam optimiser with learning rate $\text{lr} = 10^{-3}$.

Similarly to subsection 2.4, we once again use the boundary-type execution rule

$$\tau^\lambda = \inf\{t_l : V_\theta(X_{t_l}, t_l) \leq g(X_{t_l})\} \wedge T$$

to compute the value (fair price of the contract) $\mathcal{P}$ as the average

$$\mathcal{P} = \mathbb{E}\left[\delta(\tau^\lambda)g(X_{\tau^\lambda})\right],$$

where the expectation is approximated using an independent test set of 2^{18} simulated sample paths. This evaluation procedure reflects the performance of the learned randomised stopping policy when deployed as a deterministic execution boundary, and allows for a direct comparison with classical numerical benchmarks for the American put option. To ensure the reliability of the results, we repeat the experiment 100 times. For each choice of maximum stopping rate K , we aggregate the results over the 100 trials and finally report the computed value $\mathcal{P}$ and standard deviation.

Figure 8 displays the error decay for $K \in \{4, 10, 25\}$, and table 3 summarises the final values obtained after $N = 3000$ training iterations. Looking at the results, we see that the final value of the time-inconsistent American put is noticeably less than the value of the traditional time-consistent version of $\mathcal{P}_{\text{ref}} = 5.311$. Indeed, as the agent cannot commit to the globally optimal exercise strategy under time-inconsistent discounting, the equilibrium stopping rule sacrifices some option value relative to the time-consistent American put. We also see that as the maximum stopping rate K increases, the value of the option also increases slightly. This is expected, as a larger maximum rate K allows the agent to better approximate an instantaneous stopping decision, reducing the constraint imposed by bounded intensities.

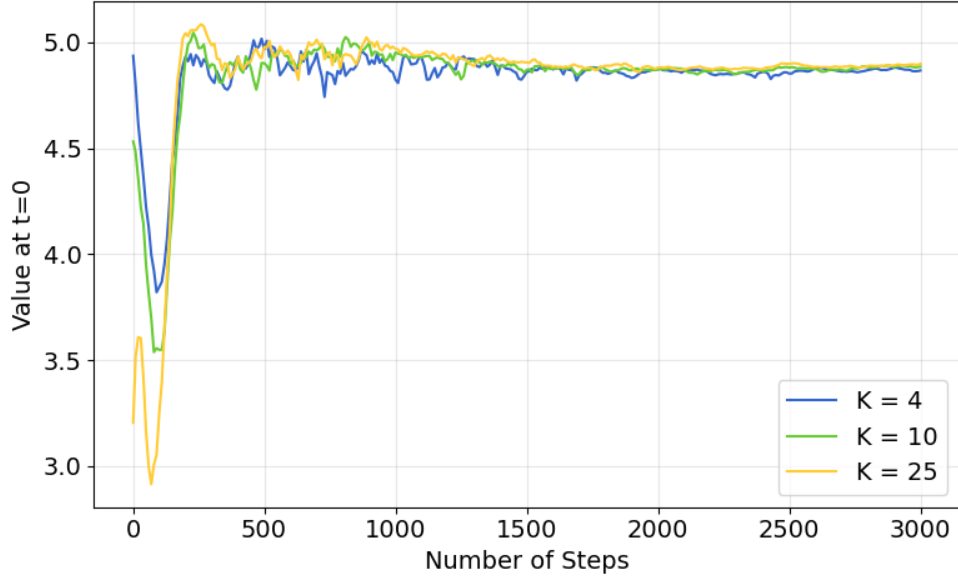


Figure 8: Learning curves for three choices of K .

K	Value $\mathcal{P}$	SD
4	4.8674	0.066
10	4.8854	0.063
25	4.8987	0.070

Table 3: Final performance of Algorithm 4 for different values of K .

6 Summary

In this thesis, we have developed a unified reinforcement learning framework for several classes of optimal stopping problems by systematically replacing the discontinuous stop/continue decision with a smooth, regularised and randomised control. The guiding principle throughout is that when the underlying state dynamics are not fully known or the classical PDE/dynamic programming methodology cannot be used, it is beneficial to introduce controlled exploration to smooth the stopping rule and ensure convergence to a solution with well-behaved policy improvement steps. This method provides clear algorithms that can be used to solve the problems numerically from simulated or observed trajectories.

In section 1, we began by exploring the classical stochastic control and optimal stopping setups, emphasising that the traditional approach to these problems typically requires the model parameters to be specified, allowing them to be solved using HJB equations and variational inequalities. This provides the conceptual baseline of the thesis: optimal stopping involves a singular decision problem with a binary action space. Here, we regularise this discontinuity at the stopping time to make it amenable to reinforcement learning while still maintaining the structure of the problem.

In section 2, we adopted the exploratory stochastic control formulation from [29], where exploration is modelled using distributions and promoted through entropy regularisation. We found that the resulting HJB equation admits a pointwise maximiser with a softmax form, which converges to the classical (pure exploitation) solution in the limit. We further noticed that adjusting the entropy regularisation parameter λ provides a clear trade-off between the exploitation of the current model and exploration of new strategies to achieve an optimal result. When applying this framework to optimal stopping we used an intensity-based approach similar to [7], determining the stopping time from an inhomogeneous Poisson process. From an algorithmic perspective this intensity formulation led naturally to a monotone policy iteration scheme, alternating between policy evaluation, which involves solving a linear PDE for the value under the current intensity, and policy improvement, where we update the intensity using a closed-form maximiser.

In section 3, we extended this approach to double stopping problems by applying the reduction principle, rewriting the setup as a single stopping problem with a reward defined through two auxiliary single stopping problems. This gave us a clean divide-and-conquer structure: train the inner single stopping problems to compute the reduced reward, then train the outer single stopping problem to determine the first exercise time, and finally recover the second stopping time by solving the remaining inner problem conditioned on the outer decision.

The final component of the thesis developed reinforcement learning methods for time-inconsistent stopping, where the non-exponential discounting factor means the ‘optimal’ decision may no longer be optimal in the future. In section 4, we solved the discrete-time decreasing impatience problem by adopting a game-theoretic perspective of stopping policies and introduced a randomised stopping rule with an entropy term to maintain exploration. This produced a one-step best-response map with a logistic closed form. In Theorem 4.1, we established monotonicity and convergence of the fixed-point iteration to an equilibrium, guaranteeing the existence of a solution. Algorithm 3 then expressed this equilibrium iteration as a simulation-based regression procedure for the continuation value.

In section 5, we explored the same problem in continuous time. We once again regularised the stopping rule through a bounded inhomogeneous Poisson intensity and derived a linear PDE for the value under a fixed candidate rate. The weak equilibrium condition reduces to a bang-bang selection rule: increase to the maximum stopping rate when the current value dominates the continuation value, or continue otherwise. This yields an equilibrium HJB system together with a monotone policy iteration scheme which converges under decreasing impatience.

Overall, the entropy-regularised viewpoint developed here provides a flexible bridge between classical optimal stopping theory and modern reinforcement learning. This thesis demonstrates that randomisation combined with entropy regularisation provides a consistent way to design stable RL algorithms across both time-consistent and time-inconsistent settings. The framework developed here has direct applications to a wide class of decision problems where optimal stopping arises under limited model knowledge. In finance, the intensity-based reinforcement learning formulation provides a robust and data-driven method to price derivatives of the American or Bermudan style. Beyond finance, the methodology applies naturally to sequential decision problems in operations research and engineering, including equipment replacement, capacity expansion and sequential testing, where stopping decisions must be made based on noisy observations.

6.1 Future Work

The framework established here suggests several natural directions for further development and extension. One important avenue is a more detailed study of the exploration parameter λ and its role in balancing bias, stability, and convergence speed. Since numerical experiments indicate a clear trade-off between faster learning and asymptotic bias, exploring how an adaptive λ could improve numerical performance is a promising direction for later research. This setup would allow us to more closely mimic simulated annealing schemes in stochastic optimisation.

Another important direction is to further investigate the gap between the PDE-level monotonicity results established in this thesis and their implementation via function approximation. The monotone policy iteration results rely heavily on exact policy evaluation via linear PDEs, whereas in practice this step is replaced by regression and parametric models as the underlying model is not known. Developing a theory of approximate monotone policy iteration by bounding projection errors or introducing constrained updates that preserve partial ordering would help bridge the gap between the analytical guarantees and algorithms used in practice.

Finally, the continuous-time time-inconsistent framework developed here raises several open questions. While the thesis establishes convergence to weak equilibrium rates under decreasing impatience, it would be interesting to characterise when weak equilibria coincide with strong or almost-strong equilibria in the entropy-regularised setting, and how regularisation selects between multiple equilibria. On the computational side, studying the stability of the rate-based policy iteration when the maximum intensity is allowed to be unbounded may clarify the connection between the randomised formulation and classical stopping regions. Collectively, these directions point toward a broader field of research in which reinforcement learning serves not only as a numerical tool, but as a unifying lens for optimal stopping and learning under uncertainty.

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